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6/17/2026



Reason (R) : A square matrix P is skew-symmetric if $P' = -P$.		
Section B (3 x 2 = 6)		
11.	If $x = e^{x/y}$, prove that $\frac{dy}{dx} = \frac{\log x - 1}{(\log x)^2}$	2

3 of 6

Q11

$$y = \log_a x$$

$$\Rightarrow x = a^y$$

illustration

$$y = 2^x$$

$$x = \log_2 y$$

$$x = e^{x/y}$$

$$\log_e e^x = \frac{x}{y}$$

$$\Rightarrow \frac{dy}{dx} = \frac{\log x - 1}{(\log x)^2}$$

Hence proved

$$z = 3^t$$

$$t = \log_3 z$$

$$\Rightarrow y = \frac{x}{\log x}$$

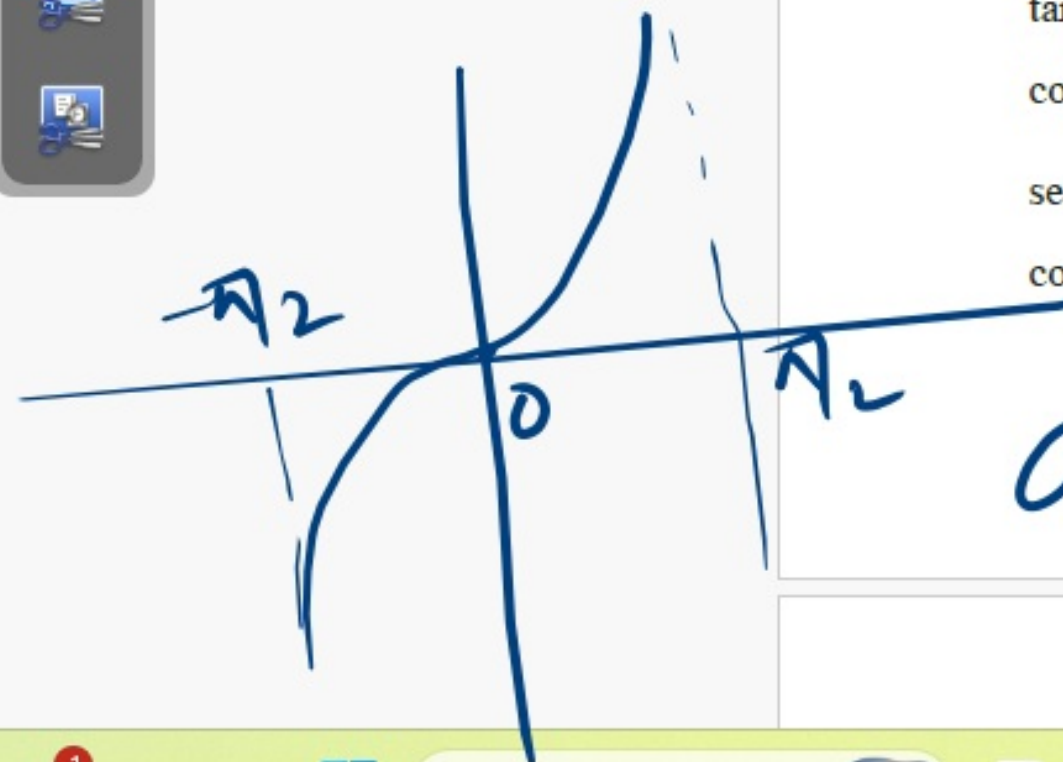
$$\Rightarrow \frac{dy}{dx} = \frac{(\log x)(1) - (x)\left(\frac{1}{x}\right)}{(\log x)^2}$$

Find the maximum and minimum value for the function $h(x) = \sin 2x + 5$ (OR) Find the absolute maximum and absolute minimum values of the function f given by $f(x) = \frac{x}{2} + \frac{2}{x}$ on the interval $[1, 2]$.	2
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 $+ 10 + 0$

fun	Domain	Rang
sin	$\mathbb{R}$	$[-1, 1]$
cos	$\mathbb{R}$	$[-1, 1]$
Tan	$\mathbb{R} - (2n+1)\frac{\pi}{2}$	$\mathbb{R}$
Cot	$\mathbb{R} - n\pi$	$\mathbb{R}$
Sec	$\mathbb{R} - (2n+1)\frac{\pi}{2}$	$\mathbb{R} - (-1, 1)$
Cosec	$\mathbb{R} - n\pi$	$\mathbb{R} - (-1, 1)$



$$y = \cot x$$

Fig 2.6 (i)

$$y = \cot^{-1} x$$

Fig 2.6 (ii)

The following table gives the inverse trigonometric function (principal value branches) along with their domains and ranges.

$\sin^{-1}$	:	$[-1, 1]$	$\rightarrow$	$\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$
$\cos^{-1}$	:	$[-1, 1]$	$\rightarrow$	$[0, \pi]$
$\operatorname{cosec}^{-1}$	:	$\mathbf{R} - (-1, 1)$	$\rightarrow$	$\left[-\frac{\pi}{2}, \frac{\pi}{2}\right] - \{0\}$
$\sec^{-1}$	:	$\mathbf{R} - (-1, 1)$	$\rightarrow$	$[0, \pi] - \left\{\frac{\pi}{2}\right\}$
$\tan^{-1}$	:	$\mathbf{R}$	$\rightarrow$	$\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$
$\cot^{-1}$	:	$\mathbf{R}$	$\rightarrow$	$(0, \pi)$

Range of Trigonometric fun
= Domain of inverse trigonometric fun^c

$y = \cot x$

Fig 2.6 (i)

$y = \cot^{-1} x$

Fig 2.6 (ii)

The following table gives the inverse trigonometric function (principal value branches) along with their domains and ranges.

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$\sec^{-1}$	:	$\mathbf{R} - (-1, 1)$	$\rightarrow$	$[0, \pi] - \left\{\frac{\pi}{2}\right\}$
$\tan^{-1}$	:	$\mathbf{R}$	$\rightarrow$	$\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$
$\cot^{-1}$	:	$\mathbf{R}$	$\rightarrow$	$(0, \pi)$

Domain

must learn

Range

Principal value branch

	Domain	Range
$\sin$	$\mathbb{R}$	$[-1, 1]$
$\cos$	$\mathbb{R}$	$[-1, 1]$
$\tan$	$\mathbb{R} - (2n+1)\frac{\pi}{2}$	$\mathbb{R}$
$\cot$	$\mathbb{R} - n\pi$	$\mathbb{R}$
$\csc$	$\mathbb{R} - n\pi$	$\mathbb{R} - (-1, 1)$
$\sec$	$\mathbb{R} - (2n+1)\frac{\pi}{2}$	$\mathbb{R} - (-1, 1)$

↓
Good to learn

	Domain	Range
$\sin^{-1}$	$[-1, 1]$	$[-\frac{\pi}{2}, \frac{\pi}{2}]$
$\cos^{-1}$	$[-1, 1]$	$[0, \pi]$
$\tan^{-1}$	$\mathbb{R}$	$(-\frac{\pi}{2}, \frac{\pi}{2})$
$\cot^{-1}$	$\mathbb{R}$	$(0, \pi) - \{\frac{\pi}{2}\}$
$\csc^{-1}$	$\mathbb{R} - (-1, 1)$	$[-\frac{\pi}{2}, \frac{\pi}{2}] - \{0\}$
$\sec^{-1}$	$\mathbb{R} - (-1, 1)$	$[0, \pi] - \{\frac{\pi}{2}\}$

↓
must learn



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Eg1

$$\sin \frac{\pi}{4} = \frac{1}{\sqrt{2}}$$

$$\frac{\pi}{4} = \sin^{-1}\left(\frac{1}{\sqrt{2}}\right)$$

$$\frac{\pi}{4} \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$$

Ans: $\sin^{-1}\left(\frac{1}{\sqrt{2}}\right) = \frac{\pi}{4}$

mean the principal value branch of that function.

3. The value of an inverse trigonometric functions which lies in the range of principal branch is called the *principal value* of that inverse trigonometric functions.

We now consider some examples:

Example 1 Find the principal value of $\sin^{-1}\left(\frac{1}{\sqrt{2}}\right)$.

Solution Let $\sin^{-1}\left(\frac{1}{\sqrt{2}}\right) = y$. Then, $\sin y = \frac{1}{\sqrt{2}}$.

We know that the range of the principal value branch of $\sin^{-1}$ is $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ and $\sin\left(\frac{\pi}{4}\right) = \frac{1}{\sqrt{2}}$. Therefore, principal value of $\sin^{-1}\left(\frac{1}{\sqrt{2}}\right)$ is $\frac{\pi}{4}$.

Example 2 Find the principal value of $\cot^{-1}\left(\frac{-1}{\sqrt{3}}\right)$.

Solution Let $\cot^{-1}\left(\frac{-1}{\sqrt{3}}\right) = y$. Then,

$$\cot y = \frac{-1}{\sqrt{3}} = -\cot\left(\frac{\pi}{3}\right) = \cot\left(\pi - \frac{\pi}{3}\right) = \cot\left(\frac{2\pi}{3}\right)$$

We know that the range of principal value branch of $\cot^{-1}$ is $(0, \pi)$ and

$\cot\left(\frac{2\pi}{3}\right) = \frac{-1}{\sqrt{3}}$. Hence, principal value of $\cot^{-1}\left(\frac{-1}{\sqrt{3}}\right)$ is $\frac{2\pi}{3}$.

EXERCISE 2.1

Find the principal values of the following:

1. $\sin^{-1}\left(-\frac{1}{2}\right)$

2. $\cos^{-1}\left(\frac{\sqrt{3}}{2}\right)$

3. $\operatorname{cosec}^{-1}(2)$

We know that the range of principal value branch of $\cot^{-1}$ is $(0, \pi)$ and
 $\cot\left(\frac{2\pi}{3}\right) = \frac{-1}{\sqrt{3}}$. Hence, principal value of $\cot^{-1}\left(\frac{-1}{\sqrt{3}}\right)$ is $\frac{2\pi}{3}$

EXERCISE 2.1

Find the principal values of the following:

- | | | |
|---|---|-----------------------------------|
| 1. $\sin^{-1}\left(-\frac{1}{2}\right)$ | 2. $\cos^{-1}\left(\frac{\sqrt{3}}{2}\right)$ | 3. $\operatorname{cosec}^{-1}(2)$ |
| 4. $\tan^{-1}(-\sqrt{3})$ | 5. $\cos^{-1}\left(-\frac{1}{2}\right)$ | 6. $\tan^{-1}(-1)$ |

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- | | | |
|---|--------------------------|--|
| 7. $\sec^{-1}\left(\frac{2}{\sqrt{3}}\right)$ | 8. $\cot^{-1}(\sqrt{3})$ | 9. $\cos^{-1}\left(-\frac{1}{\sqrt{2}}\right)$ |
| 10. $\operatorname{cosec}^{-1}(-\sqrt{2})$ | | |

Find the values of the following:

Q2

$$\cos \frac{\pi}{6} = \frac{\sqrt{3}}{2}$$

$$\Rightarrow \frac{\pi}{6} = \cos^{-1}\left(\frac{\sqrt{3}}{2}\right)$$

$$\text{as } \frac{\pi}{6} \in [0, \pi]$$

Q2

$$\cos^{-1}\left(\frac{\sqrt{3}}{2}\right)$$

$$\cos^{-1}\left(\cos \frac{\pi}{6}\right)$$

$$\frac{\pi}{6} \in [0, \pi]$$

$$\frac{\pi}{6}$$

Ans



Q6
 $\tan^{-1}(-1)$

$\tan^{-1}\left(-\tan\frac{\pi}{4}\right)$

$\tan^{-1}\left(\tan\left(\pi - \frac{\pi}{4}\right)\right)$

$\tan^{-1}\left(\tan\frac{3\pi}{4}\right)$

$\frac{3\pi}{4} \notin \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$

We know that the range of principal value branch of $\cot^{-1}$ is $(0, \pi)$ and $\cot\left(\frac{2\pi}{3}\right) = \frac{-1}{\sqrt{3}}$. Hence, principal value of $\cot^{-1}\left(\frac{-1}{\sqrt{3}}\right)$ is $\frac{2\pi}{3}$

EXERCISE 2.1

Find the principal values of the following:

1. $\sin^{-1}\left(-\frac{1}{2}\right)$
2. $\cos^{-1}\left(\frac{\sqrt{3}}{2}\right)$
3. $\operatorname{cosec}^{-1}(2)$
4. $\tan^{-1}(-\sqrt{3})$
5. $\cos^{-1}\left(-\frac{1}{2}\right)$
6. $\tan^{-1}(-1)$

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7. $\sec^{-1}\left(\frac{2}{\sqrt{3}}\right)$
 8. $\cot^{-1}(\sqrt{3})$
 9. $\cos^{-1}\left(-\frac{1}{\sqrt{2}}\right)$
 10. $\operatorname{cosec}^{-1}(-\sqrt{2})$
- Find the values of the following:

$\frac{3}{4} \times 180 = 135^\circ$
out of range

$\tan^{-1}\left(\tan\left(-\frac{\pi}{4}\right)\right)$

$-\frac{\pi}{4} \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$

$-\frac{\pi}{4}$ Ans



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Q5

$$\cos^{-1}\left(-\frac{1}{2}\right)$$

$$\cos^{-1}\left(-\cos\frac{\pi}{3}\right)$$

$$\cos^{-1}\left(\cos\left(\pi - \frac{\pi}{3}\right)\right)$$

$$\frac{2\pi}{3} \in [0, \pi]$$

Principal value

1. $\sin^{-1}\left(-\frac{1}{2}\right)$

2. $\cos^{-1}\left(\frac{\sqrt{3}}{2}\right)$

3. $\operatorname{cosec}^{-1}(2)$

Q9

$$\cos^{-1}\left(-\frac{1}{\sqrt{2}}\right)$$

4. $\tan^{-1}(-\sqrt{3})$

5. $\cos^{-1}\left(-\frac{1}{2}\right)$

6. $\tan^{-1}(-1)$

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7. $\sec^{-1}\left(\frac{2}{\sqrt{3}}\right)$

8. $\cot^{-1}(\sqrt{3})$

9. $\cos^{-1}\left(-\frac{1}{\sqrt{2}}\right)$

10. $\operatorname{cosec}^{-1}(-\sqrt{2})$

Find the values of the following:

11. $\tan^{-1}(1) + \cos^{-1}\left(-\frac{1}{2}\right) + \sin^{-1}\left(-\frac{1}{2}\right)$

12. $\cos^{-1}\left(\frac{1}{2}\right) + 2 \sin^{-1}\left(\frac{1}{2}\right)$

13. If $\sin^{-1} x = y$, then

(A) $0 \leq y \leq \pi$

(B) $-\frac{\pi}{2} \leq y \leq \frac{\pi}{2}$

H.W

Exercise 2.1 Q1 to Q14

Example - 1 and 2.

I Properties of Inverse Trigo

eg:- $\sin(\sin^{-1}(\frac{1}{2})) = \frac{1}{2}$

1) $\sin(\sin^{-1} x) = x$

this is possible when $-1 \leq x \leq 1$

2) $\cos(\cos^{-1} x) = x$

$$-1 \leq x \leq 1$$

3) $\tan(\tan^{-1} x) = x$

$$\forall x \in \mathbb{R}$$

4) $\cot(\cot^{-1} x) = x$

$$\forall x \in \mathbb{R}$$

5) $\operatorname{cosec}(\operatorname{cosec}^{-1} x) = x$

$$\forall x \in \mathbb{R} - (-1, 1)$$

6) $\sec(\sec^{-1} x) = x$

$$\forall x \in \mathbb{R} - (-1, 1)$$

II

Range

1) $\sin^{-1}(\sin x) = x$ if $x \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$

2) $\cos^{-1}(\cos x) = x$ if $x \in [0, \pi]$

3) $\tan^{-1}(\tan x) = x$ if $x \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$

4) $\cot^{-1}(\cot x) = x$ if $x \in (0, \pi)$

5) $\operatorname{cosec}^{-1}(\operatorname{cosec} x) = x$ if $x \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right] - \{0\}$

6) $\sec^{-1}(\sec x) = x$ if $x \in [0, \pi] - \{\pi/2\}$

eg: $\sin^{-1}\left(\sin \frac{12\pi}{8}\right) = \frac{12\pi}{8}$

$\sin^{-1}\left(\sin \frac{4\pi}{3}\right)$

$\sin^{-1}\left(\sin\left(\pi + \frac{\pi}{3}\right)\right)$

$\sin^{-1}\left(-\sin \frac{\pi}{3}\right)$

$\sin^{-1}\left(\sin\left(-\frac{\pi}{3}\right)\right)$

$-\frac{\pi}{3} \rightarrow \text{principal value}$

Thumb Rule / Short-Trick $\rightarrow$ ITR / C&D / Integration

- 1) $\sqrt{1-x^2}$ / $\frac{1}{\sqrt{1-x^2}}$ $x = \sin \theta$ or $\cos \theta$
- 2) $\sqrt{a^2-x^2}$ / $\frac{1}{\sqrt{a^2-x^2}}$ $x = a \sin \theta$ or $a \cos \theta$
- 3) $\sqrt{x^2-1}$ / $\frac{1}{\sqrt{x^2-1}}$ $x = \sec \theta$ or $\csc \theta$
- 4) $\sqrt{x^2+1}$ / $\frac{1}{\sqrt{x^2+1}}$ or $\sqrt{1+x^2}$ $x = \tan \theta$ or $\cot \theta$
- 5) $\sqrt{x^2+a^2}$ / $\frac{1}{\sqrt{x^2+a^2}}$ $x = a \tan \theta$ or $a \cot \theta$

$$6) \frac{\sqrt{x^2 - a^2}}{\frac{1}{\sqrt{x^2 - a^2}}}$$

$$x = a \sec \theta \text{ or } a \csc \theta$$

$$7) 3x - 4x^3 \text{ or } 4x^3 - 3x$$

$$\hookrightarrow x = \sin \theta$$

$$\hookrightarrow x = \cos \theta$$

$$8) \frac{3x - x^3}{1 - 3x^2}$$

$$\text{put } x = \tan \theta$$

$$9) \frac{3a^2x - x^3}{a^3 - 3ax^2}$$

$$\text{put } x = a \tan \theta$$

10) $\frac{1 \mp \cos x}{1 \pm \cos x}$, $\frac{1 \mp \cos x}{\sin x}$, $\frac{\sin x}{1 \mp \cos x} \rightarrow$ Always use Half angle formula.

$$\cos 2\theta = 1 - 2\sin^2 \theta$$

$$2\sin^2 \theta = 1 - \cos 2\theta$$

$$\boxed{2\sin^2 \frac{\theta}{2} = 1 - \cos \theta}$$

$$\cos 2\theta = 2\cos^2 \theta - 1$$

$$1 + \cos 2\theta = 2\cos^2 \theta$$

$$\boxed{2\cos^2 \frac{\theta}{2} = 1 + \cos \theta}$$

$$\sin 2\theta = 2\sin \theta \cos \theta$$

$$\boxed{\sin \theta = 2\sin \frac{\theta}{2} \cos \frac{\theta}{2}}$$

11) $\frac{2x}{1+x^2}$ or $\frac{1-x^2}{1+x^2}$ but $x = \tan \theta$

$\searrow \sin 2\theta$ $\searrow \cos 2\theta$

Eg³ (i)

$$\sin^{-1}(2x\sqrt{1-x^2})$$

Put $x = \sin \theta$

$$\Rightarrow \boxed{\theta = \sin^{-1} x}$$

$$\sin^{-1}(2 \sin \theta \sqrt{1-\sin^2 \theta})$$

$$\sin^{-1}(2 \sin \theta \cos \theta)$$

$$\sin^{-1}(\sin 2\theta)$$

$$2\theta = 2\sin^{-1} x$$

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We now consider some examples.

Example 3 Show that

$$(i) \sin^{-1}(2x\sqrt{1-x^2}) = 2 \sin^{-1} x, \quad -\frac{1}{\sqrt{2}} \leq x \leq \frac{1}{\sqrt{2}}$$

$$(ii) \sin^{-1}(2x\sqrt{1-x^2}) = 2 \cos^{-1} x, \quad \frac{1}{\sqrt{2}} \leq x \leq 1$$

Solution

(i) Let $x = \sin \theta$. Then $\sin^{-1} x = \theta$. We have

$$\begin{aligned} \sin^{-1}(2x\sqrt{1-x^2}) &= \sin^{-1}(2\sin \theta \sqrt{1-\sin^2 \theta}) \\ &= \sin^{-1}(2\sin \theta \cos \theta) = \sin^{-1}(\sin 2\theta) = 2\theta \\ &= 2 \sin^{-1} x \end{aligned}$$

(ii) Take $x = \cos \theta$, then proceeding as above, we get, $\sin^{-1}(2x\sqrt{1-x^2}) = 2 \cos^{-1} x$

Example 4 Express $\tan^{-1} \frac{\cos x}{1-\sin x}$, $-\frac{3\pi}{2} < x < \frac{\pi}{2}$ in the simplest form.

Solution We write

$$\tan^{-1} \left[\frac{\cos \frac{x}{2} - \sin \frac{x}{2}}{\cos \frac{x}{2} + \sin \frac{x}{2}} \right]$$

$$1 + \cot^2 \theta = \operatorname{cosec}^2 \theta$$

egs

$$x = \operatorname{cosec} \theta \Rightarrow \theta = \operatorname{cosec}^{-1} x$$

$$\cot^{-1} \left(\frac{1}{\sqrt{\operatorname{cosec}^2 \theta - 1}} \right)$$

$$\cot^{-1} \left(\frac{1}{\sqrt{\cot^2 \theta}} \right)$$

$$\cot^{-1} (\tan \theta)$$

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Example 5 Write $\cot^{-1} \left(\frac{1}{\sqrt{x^2 - 1}} \right)$, $x > 1$ in the simplest form.

Solution Let $x = \sec \theta$, then $\sqrt{x^2 - 1} = \sqrt{\sec^2 \theta - 1} = \tan \theta$

Therefore, $\cot^{-1} \frac{1}{\sqrt{x^2 - 1}} = \cot^{-1} (\cot \theta) = \theta = \sec^{-1} x$, which is the simplest form.

EXERCISE 2.2

Prove the following:

$$1. \quad 3\sin^{-1} x = \sin^{-1} (3x - 4x^3), \quad x \in \left[-\frac{1}{2}, \frac{1}{2} \right]$$

$$2. \quad 3\cos^{-1} x = \cos^{-1} (4x^3 - 3x), \quad x \in \left[\frac{1}{2}, 1 \right]$$

Write the following functions in the simplest form:

$$3. \quad \tan^{-1} \frac{\sqrt{1+x^2} - 1}{x}, \quad x \neq 0$$

$$4. \quad \tan^{-1} \left(\frac{\sqrt{1-\cos x}}{\sqrt{1+\cos x}} \right), \quad 0 < x < \pi$$

$$5. \quad \tan^{-1} (\cos x - \sin x), \quad -\pi < x < 3\pi$$

$$x = \sec \theta \Rightarrow \theta = \sec^{-1} x$$

$$\cot^{-1} \left(\frac{1}{\sqrt{\sec^2 \theta - 1}} \right)$$

$$\cot^{-1} \left(\frac{1}{\tan \theta} \right)$$

$$\cot^{-1} (\cot \theta)$$

$$\sec^{-1} x$$

Q6

$$\tan^{-1} \left(\frac{x}{\sqrt{a^2 - x^2}} \right)$$

put $x = a \sin \theta$

$$\Rightarrow \frac{x}{a} = \sin \theta$$

$$\Rightarrow \theta = \sin^{-1} \left(\frac{x}{a} \right)$$

$$\tan^{-1} \left(\frac{a \sin \theta}{\sqrt{a^2 - a^2 \sin^2 \theta}} \right)$$

$$\tan^{-1} \left(\frac{a \sin \theta}{\sqrt{a^2 (1 - \sin^2 \theta)}} \right)$$

$$1. \quad 3 \sin^{-1} x = \sin^{-1} (3x - 4x^3), \quad x \in \left[-\frac{1}{2}, \frac{1}{2} \right]$$

$$2. \quad 3 \cos^{-1} x = \cos^{-1} (4x^3 - 3x), \quad x \in \left[\frac{1}{2}, 1 \right]$$

Write the following functions in the simplest form:

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$$4. \quad \tan^{-1} \left(\sqrt{\frac{1-\cos x}{1+\cos x}} \right), \quad 0 < x < \pi$$

$$5. \quad \tan^{-1} \left(\frac{\cos x - \sin x}{\cos x + \sin x} \right), \quad -\frac{\pi}{4} < x < \frac{3\pi}{4}$$

$$6. \quad \tan^{-1} \frac{x}{\sqrt{a^2 - x^2}}, \quad |x| < a$$

$$7. \quad \tan^{-1} \left(\frac{3a^2 x - x^3}{a^3 - 3ax^2} \right), \quad a > 0; \quad \frac{-a}{\sqrt{3}} < x < \frac{a}{\sqrt{3}}$$

Find the values of each of the following:

$$8. \quad \tan^{-1} \left[2 \cos \left(2 \sin^{-1} \frac{1}{2} \right) \right]$$

$$9. \quad \tan \frac{1}{2} \left[\sin^{-1} \frac{2x}{1+x^2} + \cos^{-1} \frac{1-y^2}{1+y^2} \right], \quad |x| < 1, y > 0 \text{ and } xy < 1$$

$$\tan \left(\frac{a \sin \theta}{a \cos \theta} \right)$$

$$\tan^{-1} (\tan \theta)$$

$$\theta = \sin^{-1} \left(\frac{x}{a} \right)$$

$$\theta = \cot^{-1} \left(\frac{\sqrt{a^2 - x^2}}{x} \right)$$

Therefore, $\cot^{-1} \frac{1}{\sqrt{x^2-1}} = \cot^{-1}(\cot \theta) = \theta = \sec^{-1} x$, which is the simplest form.

EXERCISE 2.2

Prove the following:

1. $3\sin^{-1} x = \sin^{-1}(3x - 4x^3)$, $x \in \left[-\frac{1}{2}, \frac{1}{2}\right]$

2. $3\cos^{-1} x = \cos^{-1}(4x^3 - 3x)$, $x \in \left[\frac{1}{2}, 1\right]$

Write the following functions in the simplest form:

3. $\tan^{-1} \frac{\sqrt{1+x^2}-1}{x}$, $x \neq 0$

4. $\tan^{-1} \left(\sqrt{\frac{1-\cos x}{1+\cos x}} \right)$, $0 < x < \pi$

5. $\tan^{-1} \left(\frac{\cos x - \sin x}{\cos x + \sin x} \right)$, $-\frac{\pi}{4} < x < \frac{3\pi}{4}$

6. $\tan^{-1} \frac{x}{\sqrt{a^2-x^2}}$, $|x| < a$

7. $\tan^{-1} \left(\frac{3a^2x - x^3}{a^3 - 3ax^2} \right)$, $a > 0$; $-\frac{a}{\sqrt{3}} < x < \frac{a}{\sqrt{3}}$

Find the values of each of the following:

8. $\tan^{-1} \left[2 \cos \left(2 \sin^{-1} \frac{1}{2} \right) \right]$

9. $\tan^{-1} \frac{1}{2} \left[\sin^{-1} \frac{2x}{1+x^2} + \cos^{-1} \frac{1-y^2}{1+y^2} \right]$, $|x| < 1$, $y > 0$ and $xy < 1$

Q1
RHS

$$\sin^{-1}(3x - 4x^3)$$

$$x = \sin \theta$$

$$\Rightarrow \theta = \sin^{-1} x$$

$$\sin^{-1}(3 \sin \theta - 4 \sin^3 \theta)$$

$$\sin^{-1}(\sin 3\theta)$$

$$3\theta$$

$$3 \sin^{-1} x = \text{LHS}$$

Simple form

Q7

$$\tan^{-1} \left(\frac{3a^2x - x^3}{a^3 - 3ax^2} \right)$$

$$\text{Put } x = a \tan \theta$$

$$\Rightarrow \frac{x}{a} = \tan \theta$$

$$\Rightarrow \theta = \tan^{-1} \left(\frac{x}{a} \right)$$

$$\tan^{-1} \left(\frac{3a^2(a \tan \theta) - a^3 \tan^3 \theta}{a^3 - 3a(a^2 \tan^2 \theta)} \right)$$

$$2. \quad 3\cos^{-1} x = \cos^{-1} (4x^3 - 3x), \quad x \in \left[\frac{1}{2}, 1 \right]$$

Write the following functions in the simplest form:

$$3. \quad \tan^{-1} \frac{\sqrt{1+x^2} - 1}{x}, \quad x \neq 0$$

$$5. \quad \tan^{-1} \left(\frac{\cos x - \sin x}{\cos x + \sin x} \right), \quad -\frac{\pi}{4} < x < \frac{3\pi}{4}$$

$$6. \quad \tan^{-1} \frac{x}{\sqrt{a^2 - x^2}}, \quad |x| < a$$

$$7. \quad \tan^{-1} \left(\frac{3a^2x - x^3}{a^3 - 3ax^2} \right), \quad a > 0; \quad \frac{-a}{\sqrt{3}} < x < \frac{a}{\sqrt{3}}$$

Find the values of each of the following:

$$8. \quad \tan^{-1} \left[2 \cos \left(2 \sin^{-1} \frac{1}{2} \right) \right]$$

$$9. \quad \tan^{-1} \left[\sin^{-1} \frac{2x}{1+x^2} + \cos^{-1} \frac{1-y^2}{1+y^2} \right], \quad |x| < 1, y > 0 \text{ and } xy < 1$$

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$$\text{find } \frac{dy}{dx} \quad \tan^{-1} \left(\sqrt{\frac{1-\cos x}{1+\cos x}} \right), \quad 0 < x < \pi$$

$$\tan^{-1} \left(\frac{a^3 (3 \tan \theta - \tan^3 \theta)}{a^3 (1 - 3 \tan^2 \theta)} \right)$$

$$\tan^{-1} (\tan 3\theta)$$

$$3\theta$$

$$3 \tan^{-1} \left(\frac{x}{a} \right)$$

$$y =$$

$$\frac{dy}{dx} = 3 \cdot \frac{1}{1 + \left(\frac{x}{a} \right)^2} \cdot \left(\frac{1}{a} \right)$$

$$= 3 \cdot \frac{1}{\frac{a^2 + x^2}{a^2}} \cdot \frac{1}{a}$$

$$\left(\tan^{-1} x \right)' = \frac{1}{1+x^2}$$

$$= 3 \cdot \frac{a}{a^2 + x^2} \cdot \frac{1}{a}$$

$$\text{Q4} \quad \tan^{-1} \left(\sqrt{\frac{1 - \cos x}{1 + \cos x}} \right)$$

$$\tan^{-1} \left(\sqrt{\frac{2 \sin^2 \frac{x}{2}}{2 \cos^2 \frac{x}{2}}} \right)$$

$$\tan^{-1} \left(\sqrt{\tan^2 \frac{x}{2}} \right)$$

$$\tan^{-1} \left(\tan \frac{x}{2} \right)$$

$$\frac{x}{2}$$

$$2. \quad 3 \cos^{-1} x = \cos^{-1} (4x^3 - 3x), \quad x \in \left[-\frac{1}{2}, 1 \right]$$

Write the following functions in the simplest form:

$$3. \quad \tan^{-1} \frac{\sqrt{1+x^2}-1}{x}, \quad x \neq 0$$

$$4. \quad \tan^{-1} \left(\sqrt{\frac{1-\cos x}{1+\cos x}} \right), \quad 0 < x < \pi$$

$$5. \quad \tan^{-1} \left(\frac{\cos x - \sin x}{\cos x + \sin x} \right), \quad -\frac{\pi}{4} < x < \frac{3\pi}{4}$$

$$6. \quad \tan^{-1} \frac{x}{\sqrt{a^2 - x^2}}, \quad |x| < a$$

$$7. \quad \tan^{-1} \left(\frac{3a^2x - x^3}{a^3 - 3ax^2} \right), \quad a > 0; \quad -\frac{a}{\sqrt{3}} < x < \frac{a}{\sqrt{3}}$$

Find the values of each of the following:

$$8. \quad \tan^{-1} \left[2 \cos \left(2 \sin^{-1} \frac{1}{2} \right) \right]$$

$$9. \quad \tan \frac{1}{2} \left[\sin^{-1} \frac{2x}{1+x^2} + \cos^{-1} \frac{1-y^2}{1+y^2} \right], \quad |x| < 1, y > 0 \text{ and } xy < 1$$

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$$\text{Q5} \quad \tan^{-1} \left(\frac{\cos x - \sin x}{\cos x + \sin x} \right)$$

Divide by $\cos x$ in N^r ,

$$\tan^{-1} \left(\frac{1 - \tan x}{1 + 1 \cdot \tan x} \right)$$

$$\tan^{-1} \left(\frac{\tan \frac{\pi}{4} - \tan x}{1 + \tan \frac{\pi}{4} \tan x} \right)$$

$$\tan^{-1} \left(\tan \left(\frac{\pi}{4} - x \right) \right)$$

$$\frac{\pi}{4} - x$$

Q

$$\tan^{-1} \left(\frac{\cos x + \sin x}{\cos x - \sin x} \right)$$

Eg 4

$$\tan^{-1} \left(\frac{\cos x}{1 - \sin x} \right)$$

$$\tan^{-1} \left(\frac{\sin \left(\frac{\pi}{2} - x \right)}{1 - \cos \left(\frac{\pi}{2} - x \right)} \right)$$

$$\sin \theta = 2 \sin \frac{\theta}{2} \cos \frac{\theta}{2}$$

$$1 - \cos \theta = 2 \sin^2 \frac{\theta}{2}$$

$$\tan^{-1} \left(\frac{2 \sin \left(\frac{\pi}{4} - \frac{x}{2} \right) \cos \left(\frac{\pi}{4} - \frac{x}{2} \right)}{2 \sin^2 \left(\frac{\pi}{4} - \frac{x}{2} \right)} \right)$$

(i) Let $x = \sin \theta$. Then $\sin^{-1} x = \theta$. We have

$$\begin{aligned} \sin^{-1} (2x\sqrt{1-x^2}) &= \sin^{-1} (2 \sin \theta \sqrt{1 - \sin^2 \theta}) \\ &= \sin^{-1} (2 \sin \theta \cos \theta) = \sin^{-1} (\sin 2\theta) = 2\theta \\ &= 2 \sin^{-1} x \end{aligned}$$

(ii) Take $x = \cos \theta$, then proceeding as above, we get, $\sin^{-1} (2x\sqrt{1-x^2}) = 2 \cos^{-1} x$

Example 4 Express $\tan^{-1} \frac{\cos x}{1 - \sin x}$, $-\frac{3\pi}{2} < x < \frac{\pi}{2}$ in the simplest form.

Solution We write

$$\tan^{-1} \left(\frac{\cos x}{1 - \sin x} \right) = \tan^{-1} \left[\frac{\cos^2 \frac{x}{2} - \sin^2 \frac{x}{2}}{\cos^2 \frac{x}{2} + \sin^2 \frac{x}{2} - 2 \sin \frac{x}{2} \cos \frac{x}{2}} \right]$$

$$= \tan^{-1} \left[\frac{\left(\cos \frac{x}{2} + \sin \frac{x}{2} \right) \left(\cos \frac{x}{2} - \sin \frac{x}{2} \right)}{\left(\cos \frac{x}{2} - \sin \frac{x}{2} \right)^2} \right]$$

$$= \tan^{-1} \left[\frac{\cos \frac{x}{2} + \sin \frac{x}{2}}{\cos \frac{x}{2} - \sin \frac{x}{2}} \right] = \tan^{-1} \left[\frac{1 + \tan \frac{x}{2}}{1 - \tan \frac{x}{2}} \right]$$

$$= \tan^{-1} \left[\tan \left(\frac{\pi}{4} + \frac{x}{2} \right) \right] = \frac{\pi}{4} + \frac{x}{2}$$

$$\tan^{-1} \left(\frac{\cos \left(\frac{\pi}{4} - \frac{x}{2} \right)}{\sin \left(\frac{\pi}{4} - \frac{x}{2} \right)} \right)$$

$$\tan^{-1} \left(\cot \left(\frac{\pi}{4} - \frac{x}{2} \right) \right)$$

$$\tan \left(\frac{\pi}{2} - \theta \right) = \cot \theta$$

$$\tan^{-1} \left(\tan \left(\frac{\pi}{2} - \left(\frac{\pi}{4} - \frac{x}{2} \right) \right) \right)$$

$$\frac{\pi}{2} - \frac{\pi}{4} + \frac{x}{2}$$

$$\frac{\pi}{4} + \frac{x}{2}$$

Q $\cot^{-1} \left(\frac{1 + \sin x}{\cos x} \right)$

$$\cot^{-1} \left(\frac{1 + \cos \left(\frac{\pi}{2} - x \right)}{\sin \left(\frac{\pi}{2} - x \right)} \right)$$

$$\cot^{-1} \left(\frac{2 \cos^2 \left(\frac{\pi}{4} - \frac{x}{2} \right)}{2 \sin \left(\frac{\pi}{4} - \frac{x}{2} \right) \cos \left(\frac{\pi}{4} - \frac{x}{2} \right)} \right)$$

$$\cot^{-1} \left(\frac{\cos \left(\frac{\pi}{4} - \frac{x}{2} \right)}{\sin \left(\frac{\pi}{4} - \frac{x}{2} \right)} \right)$$

$$\cot^{-1} \left(\cot \left(\frac{\pi}{4} - \frac{x}{2} \right) \right)$$

$$\frac{\pi}{4} - \frac{x}{2}$$

09

Let $x = \tan \alpha$ and $y = \tan \beta$
 $\Rightarrow \alpha = \tan^{-1} x$ and $\beta = \tan^{-1} y$

$$\tan \frac{1}{2} \left[\sin^{-1} \left(\frac{2 \tan \alpha}{1 + \tan^2 \alpha} \right) \right]$$

$$\tan \frac{1}{2} \left[\sin^{-1} (\sin^2 \alpha) + \cos^{-1} (\cos^2 \alpha) \right]$$

$$5. \tan^{-1} \left(\frac{\cos x - \sin x}{\cos x + \sin x} \right), \frac{-\pi}{4} < x < \frac{3\pi}{4}$$

$$6. \tan^{-1} \frac{x}{\sqrt{a^2 - x^2}}, |x| < a$$

$$7. \tan^{-1} \left(\frac{3a^2 x - x^3}{a^3 - 3ax^2} \right), a > 0; \frac{-a}{\sqrt{3}} < x < \frac{a}{\sqrt{3}}$$

Find the values of each of the following:

$$8. \tan^{-1} \left[2 \cos \left(2 \sin^{-1} \frac{1}{2} \right) \right]$$

$$9. \tan \frac{1}{2} \left[\sin^{-1} \frac{2x}{1+x^2} + \cos^{-1} \frac{1-y^2}{1+y^2} \right], |x| < 1, y > 0 \text{ and } xy < 1$$

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$$\tan \frac{1}{2} [2\alpha + 2\beta]$$

$$\tan (\alpha + \beta)$$

$$\frac{\tan \alpha + \tan \beta}{1 - \tan \alpha \tan \beta}$$

$$\left[\frac{x + y}{1 - xy} \right]$$

Ans

H.W

Example - 3, 4, 5

Exercise 2.2 - Q1 to Q7 + Q9

Misc exercise - Q9, Q10, Q12

Monday

Q10

$$\sin^{-1}\left(\sin \frac{2\pi}{3}\right)$$

$$= \frac{2\pi}{3} \notin \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$$

↳ Not possible.

$$\sin^{-1}\left(\sin\left(\pi - \frac{\pi}{3}\right)\right)$$

$$\sin^{-1}\left(\sin \frac{\pi}{3}\right)$$

$$\frac{\pi}{3} \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$$

↳ Ans.

Find the values of each of the following:

8. $\tan^{-1}\left[2 \cos\left(2 \sin^{-1} \frac{1}{2}\right)\right]$

9. $\tan \frac{1}{2}\left[\sin^{-1} \frac{2x}{1+x^2} + \cos^{-1} \frac{1-y^2}{1+y^2}\right], |x| < 1, y > 0 \text{ and } xy < 1$

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Rough

$$\frac{2\pi}{3} = \frac{2}{3} \times 180 = 120^\circ$$

Q11 $\tan^{-1}\left(\tan \frac{3\pi}{4}\right)$

$$-\frac{\pi}{4} \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$$

↳ Ans

30 MATHEMATICS

Find the values of each of the expressions in Exercises 16 to 18.

10. $\sin^{-1}\left(\sin \frac{2\pi}{3}\right)$

11. $\tan^{-1}\left(\tan \frac{3\pi}{4}\right)$

12. $\tan\left(\sin^{-1} \frac{3}{5} + \cot^{-1} \frac{3}{2}\right)$

13. $\cos^{-1}\left(\cos \frac{7\pi}{4}\right)$ is equal to

Q11

$$\tan^{-1}\left(\tan\left(\pi - \frac{\pi}{4}\right)\right)$$

$$\tan^{-1}\left(-\tan\left(\frac{\pi}{4}\right)\right)$$

$$\tan^{-1}\left(\tan\left(-\frac{\pi}{4}\right)\right)$$

$$-\frac{\pi}{4} \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$$

→ Ans

30 MATHEMATICS

Find the values of each of the expressions in Exercises 16 to 18.

10. $\sin^{-1}\left(\sin \frac{2\pi}{3}\right)$ 11. $\tan^{-1}\left(\tan \frac{3\pi}{4}\right)$
12. $\tan\left(\sin^{-1} \frac{3}{5} + \cot^{-1} \frac{3}{2}\right)$
13. $\cos^{-1}\left(\cos \frac{7\pi}{6}\right)$ is equal to
 (A) $\frac{7\pi}{6}$ (B) $\frac{5\pi}{6}$ (C) $\frac{\pi}{3}$ (D) $\frac{\pi}{6}$
14. $\sin\left(\frac{\pi}{3} - \sin^{-1}\left(-\frac{1}{2}\right)\right)$ is equal to
 (A) $\frac{1}{2}$ (B) $\frac{1}{3}$ (C) $\frac{1}{4}$ (D) 1
15. $\tan^{-1} \sqrt{3} - \cot^{-1}(-\sqrt{3})$ is equal to
 (A) π (B) $-\frac{\pi}{2}$ (C) 0 (D) $2\sqrt{3}$

Miscellaneous Examples

Example 6 Find the value of $\sin^{-1}\left(\sin \frac{3\pi}{5}\right)$

Solution We know that $\sin^{-1}(\sin x) = x$. Therefore $\sin^{-1}\left(\sin \frac{3\pi}{5}\right) = \frac{3\pi}{5}$

Q13

$$\cos^{-1}\left(\cos \frac{7\pi}{6}\right) \neq \frac{7\pi}{6} \rightarrow \text{out of range}$$

$$\cos^{-1}\left(\cos\left(\pi + \frac{\pi}{6}\right)\right)$$

$$\cos^{-1}\left(-\cos \frac{\pi}{6}\right)$$

$$\cos^{-1}\left(\cos\left(\pi - \frac{\pi}{6}\right)\right)$$

$$\frac{5\pi}{6} \in [0, \pi]$$

→ Ans.



Eg6 $\sin^{-1}\left(\sin \frac{3\pi}{5}\right)$

$\sin^{-1}\left(\sin\left(\pi - \frac{3\pi}{5}\right)\right)$

$\frac{2\pi}{5} \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$

→ No

$\tan^{-1}\left(\tan \frac{5\pi}{8}\right)$

Find the values of each of the expressions in Exercises 16 to 18.

- 10. $\sin^{-1}\left(\sin \frac{2\pi}{3}\right)$
- 11. $\tan^{-1}\left(\tan \frac{3\pi}{4}\right)$
- 12. $\tan\left(\sin^{-1} \frac{3}{5} + \cot^{-1} \frac{3}{2}\right)$
- 13. $\cos^{-1}\left(\cos \frac{7\pi}{6}\right)$ is equal to
(A) $\frac{7\pi}{6}$ (B) $\frac{5\pi}{6}$ (C) $\frac{\pi}{3}$ (D) $\frac{\pi}{6}$
- 14. $\sin\left(\frac{\pi}{3} - \sin^{-1}\left(-\frac{1}{2}\right)\right)$ is equal to
(A) $\frac{1}{2}$ (B) $\frac{1}{3}$ (C) $\frac{1}{4}$ (D) 1
- 15. $\tan^{-1} \sqrt{3} - \cot^{-1}(-\sqrt{3})$ is equal to
(A) π (B) $-\frac{\pi}{2}$ (C) 0 (D) $2\sqrt{3}$

Miscellaneous Examples

Example 6 Find the value of $\sin^{-1}\left(\sin \frac{3\pi}{5}\right)$

$\frac{2\pi}{5} = \frac{2}{5} \times 180 = 72^\circ$

$\frac{3\pi}{5} = \frac{3}{5} \times 180 = 108^\circ$
out of range

∴ Logic

$\text{I} \quad \left(\frac{\pi - 0}{\pi}\right)$	$\text{II} \quad (0)$
$\text{III} \quad \left(\frac{\pi + 0}{\pi}\right)$	$\text{IV} \quad (2\pi - 0)$



$\tan^{-1} \left(\tan \left(-\pi + \frac{5\pi}{8} \right) \right)$

$\tan^{-1} \left(\tan \left(-\frac{3\pi}{8} \right) \right)$

$-\frac{3\pi}{8} \in \left(-\frac{\pi}{2}, \frac{\pi}{2} \right)$
Ans.

$\neq \frac{5\pi}{8}$
 $\tan^{-1} \left(\tan \frac{5\pi}{8} \right)$

30 MATHEMATICS

Find the values of each of the expressions in Exercises 16 to 18.

10. $\sin^{-1} \left(\sin \frac{2\pi}{3} \right)$

11. $\tan^{-1} \left(\tan \frac{3\pi}{4} \right)$

12. $\tan \left(\sin^{-1} \frac{3}{5} + \cot^{-1} \frac{3}{2} \right)$

$\cos^{-1} \left(\cos \frac{7\pi}{6} \right)$ is equal to

(A) $\frac{7\pi}{6}$

(B) $\frac{5\pi}{6}$

(C) $\frac{\pi}{3}$

(D) $\frac{\pi}{6}$

14. $\sin \left(\frac{\pi}{3} - \sin^{-1} \left(-\frac{1}{2} \right) \right)$ is equal to

(A) $\frac{1}{2}$

(B) $\frac{1}{3}$

(C) $\frac{1}{4}$

(D) 1

15. $\tan^{-1} \sqrt{3} - \cot^{-1} (-\sqrt{3})$ is equal to

(A) π

(B) $-\frac{\pi}{2}$

(C) 0

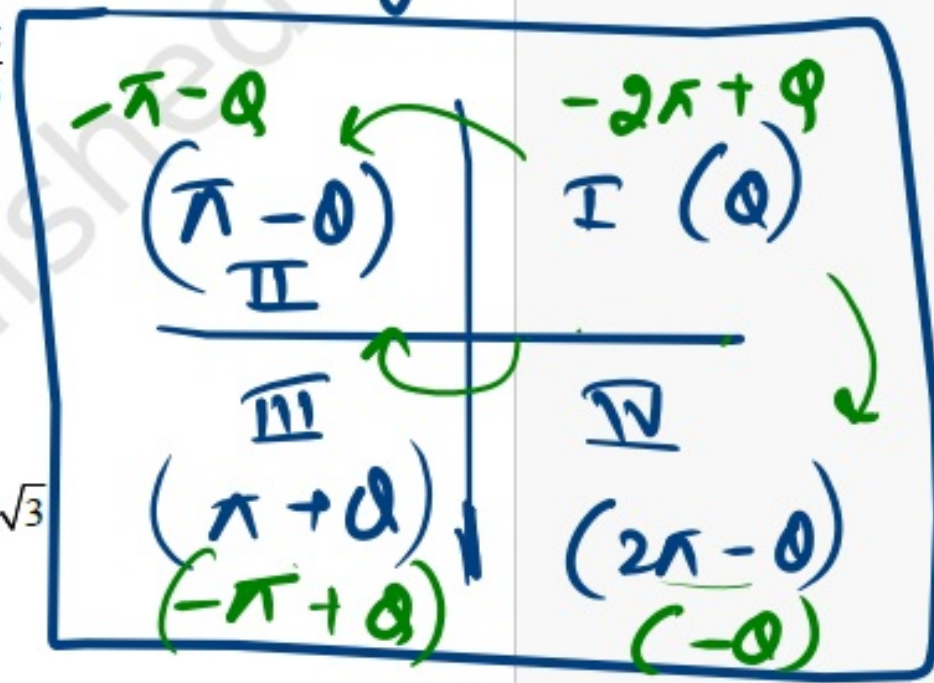
(D) $2\sqrt{3}$

Miscellaneous Examples

Example 6 Find the value of $\sin^{-1} \left(\sin \frac{3\pi}{5} \right)$

$\frac{5}{8} \times 180 = 112.5$
 $-\frac{3\pi}{8} = -\frac{3}{8} \times 180 = -135$

Logic





Q15

$$\tan^{-1}(\sqrt{3}) - \cot^{-1}(-\sqrt{3})$$

MATHEMATICS

$$\tan^{-1}\left(\tan \frac{\pi}{3}\right)$$

$$\frac{\pi}{3}$$

$$\frac{\pi}{3} - \frac{5\pi}{6}$$

$$\frac{2\pi - 5\pi}{6} = -\frac{3\pi}{6} = -\frac{\pi}{2}$$

Find the values of each of the expressions in Exercises 16 to 18.

10. $\sin^{-1}\left(\sin \frac{2\pi}{3}\right)$

11. $\tan^{-1}\left(\tan \frac{3\pi}{4}\right)$

12. $\tan\left(\sin^{-1} \frac{3}{5} + \cot^{-1} \frac{3}{2}\right)$

13. $\cos^{-1}\left(\cos \frac{7\pi}{6}\right)$ is equal to

(A) $\frac{7\pi}{6}$

(B) $\frac{5\pi}{6}$

(C) $\frac{\pi}{3}$

(D) $\frac{\pi}{6}$

14. $\sin\left(\frac{\pi}{3} - \sin^{-1}\left(-\frac{1}{2}\right)\right)$ is equal to

(A) $\frac{1}{2}$

(B) $\frac{1}{3}$

(C) $\frac{1}{4}$

(D) $\frac{1}{5}$

15. $\tan^{-1} \sqrt{3} - \cot^{-1}(-\sqrt{3})$ is equal to

(A) π

(B) $-\frac{\pi}{2}$

(C) 0

(D) $2\sqrt{3}$

Miscellaneous Examples

Example 6 Find the value of $\sin^{-1}\left(\sin \frac{3\pi}{5}\right)$

$$\cot^{-1}(-\sqrt{3})$$

$$\cot^{-1}\left(-\cot \frac{\pi}{6}\right)$$

$$\cot^{-1}\left(\cot\left(\pi - \frac{\pi}{6}\right)\right)$$

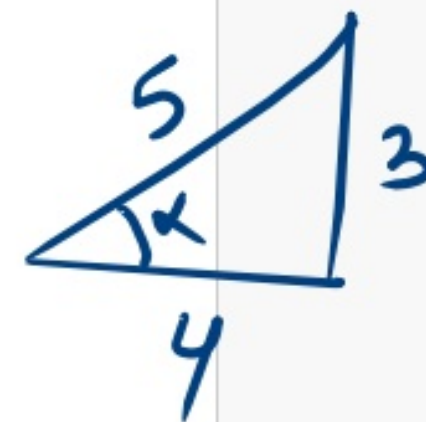
$$\frac{5\pi}{6} \in (0, \pi)$$

$-\pi - \theta$	$-2\pi + \theta$
$\pi - \theta$	θ
$\pi + \theta$	$2\pi - \theta$
$-\pi + \theta$	$-\theta$

concept

Final Ans: $\frac{25}{6}$

$$\sin^{-1} \frac{3}{5} = \alpha \Rightarrow \frac{3}{5} = \sin \alpha$$



$$\tan \alpha = \frac{3}{4}$$

$$\cot^{-1} \frac{3}{2} = \beta \Rightarrow \frac{3}{2} = \cot \beta$$

$$\Rightarrow \tan \beta = \frac{1}{\cot \beta} = \frac{2}{3}$$

$$\tan(\alpha + \beta) = \frac{\frac{3}{4} + \frac{2}{3}}{1 - \frac{3}{4} \times \frac{2}{3}} = \frac{\frac{25}{12}}{\frac{1}{2}}$$

$$\tan(\alpha + \beta) = \frac{25}{6}$$

Q12

$$\tan\left(\sin^{-1} \frac{3}{5} + \cot^{-1} \frac{3}{2}\right)$$

$$\text{Let } \sin^{-1} \frac{3}{5} = \alpha$$

$$\text{and } \cot^{-1} \frac{3}{2} = \beta$$

$$\tan(\alpha + \beta) = \frac{\tan \alpha + \tan \beta}{1 - \tan \alpha \tan \beta}$$

Find the values of each of the expressions in Exercises 16 to 18.

10. $\sin^{-1}\left(\sin \frac{2\pi}{3}\right)$

11. $\tan^{-1}\left(\tan \frac{3\pi}{4}\right)$

12. $\tan\left(\sin^{-1} \frac{3}{5} + \cot^{-1} \frac{3}{2}\right)$

13. $\cos^{-1}\left(\cos \frac{7\pi}{6}\right)$ is equal to

(A) $\frac{7\pi}{6}$ (B) $\frac{5\pi}{6}$ (C) $\frac{\pi}{3}$ (D) $\frac{\pi}{6}$

$\tan^{-1}\left(\frac{1}{3}\right) - \sin^{-1}\left(\frac{1}{2}\right)$ is equal to

(A) $\frac{1}{2}$ (B) $\frac{1}{3}$ (C) $\frac{1}{4}$ (D) 1

15. $\tan^{-1} \sqrt{3} - \cot^{-1}(-\sqrt{3})$ is equal to

(A) π (B) $-\frac{\pi}{2}$ (C) 0 (D) $2\sqrt{3}$

INVERSE TRIGONOMETRIC FUNCTIONS 31

Miscellaneous Exercise on Chapter 2

Find the value of the following:

1. $\cos^{-1}\left(\cos\frac{13\pi}{6}\right)$

2. $\tan^{-1}\left(\tan\frac{7\pi}{6}\right)$

3. $2\sin^{-1}\frac{3}{5} = \tan^{-1}\frac{24}{7}$

4. $\sin^{-1}\frac{8}{17} + \sin^{-1}\frac{3}{5} = \tan^{-1}\frac{77}{36}$

5. $\cos^{-1}\frac{4}{5} + \cos^{-1}\frac{12}{13} = \cos^{-1}\frac{33}{65}$

6. $\cos^{-1}\frac{12}{13} + \sin^{-1}\frac{3}{5} = \sin^{-1}\frac{56}{65}$

7. $\tan^{-1}\frac{63}{16} = \sin^{-1}\frac{5}{13} + \cos^{-1}\frac{3}{5}$

Prove that

8. $\tan^{-1}\sqrt{x} = \frac{1}{2}\cos^{-1}\frac{1-x}{1+x}, x \in [0, 1]$

9. $\cot^{-1}\left(\frac{\sqrt{1+\sin x} + \sqrt{1-\sin x}}{\sqrt{1+\sin x} - \sqrt{1-\sin x}}\right) = \frac{x}{2}, x \in \left(0, \frac{\pi}{4}\right)$

10. $\tan^{-1}\left(\frac{\sqrt{1+x} - \sqrt{1-x}}{\sqrt{1+x} + \sqrt{1-x}}\right) = \frac{\pi}{4} - \frac{1}{2}\cos^{-1}x, -\frac{1}{\sqrt{2}} \leq x \leq 1$ [Hint: Put $x = \cos 2\theta$]

Solve the following equations:

$\tan^{-1}x + \tan^{-1}y = \tan^{-1}\left(\frac{x+y}{1-xy}\right)$

Not in course

removed

Q4

$\sin^{-1}\frac{8}{17} + \sin^{-1}\frac{3}{5} = \tan^{-1}\frac{77}{36}$

$\Rightarrow \tan\left(\sin^{-1}\frac{8}{17} + \sin^{-1}\frac{3}{5}\right) = \frac{77}{36}$

How to solve this



Miscellaneous Exercise on Chapter 2

Find the value of the following:

1. $\cos^{-1}\left(\cos\frac{13\pi}{6}\right)$

2. $\tan^{-1}\left(\tan\frac{7\pi}{6}\right)$

Prove that

3. $2\sin^{-1}\frac{3}{5} = \tan^{-1}\frac{24}{7}$

4. $\sin^{-1}\frac{8}{17} + \sin^{-1}\frac{3}{5} = \tan^{-1}\frac{77}{36}$

5. $\cos^{-1}\frac{4}{5} + \cos^{-1}\frac{12}{13} = \cos^{-1}\frac{33}{65}$

6. $\cos^{-1}\frac{12}{13} + \sin^{-1}\frac{3}{5} = \sin^{-1}\frac{56}{65}$

7. $\tan^{-1}\frac{63}{16} = \sin^{-1}\frac{5}{13} + \cos^{-1}\frac{3}{5}$

Prove that

8. $\tan^{-1}\sqrt{x} = \frac{1}{2}\cos^{-1}\frac{1-x}{1+x}, x \in [0, 1]$

9. $\cot^{-1}\left(\frac{\sqrt{1+\sin x} + \sqrt{1-\sin x}}{\sqrt{1+\sin x} - \sqrt{1-\sin x}}\right) = \frac{x}{2}, x \in \left(0, \frac{\pi}{4}\right)$

10. $\tan^{-1}\left(\frac{\sqrt{1+x} - \sqrt{1-x}}{\sqrt{1+x} + \sqrt{1-x}}\right) = \frac{\pi}{4} - \frac{1}{2}\cos^{-1}x, -\frac{1}{\sqrt{2}} \leq x \leq \frac{1}{\sqrt{2}}$ [Hint: Put $x = \cos 2\theta$]

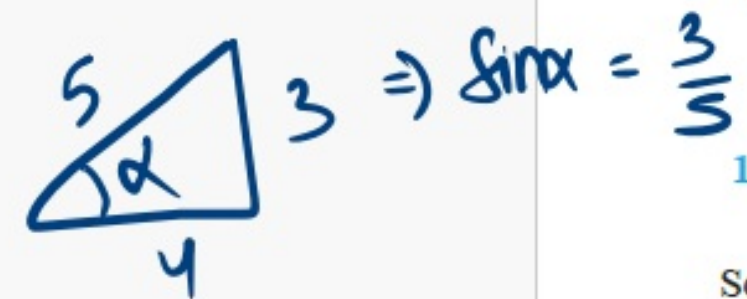
Solve the following equations:

Q5
 $\cos\left(\underbrace{\cos^{-1}\frac{4}{5}}_{\alpha} + \underbrace{\cos^{-1}\frac{12}{13}}_{\beta}\right) = \frac{33}{65}$

$\cos(\alpha + \beta) = \cos\alpha\cos\beta - \sin\alpha\sin\beta$

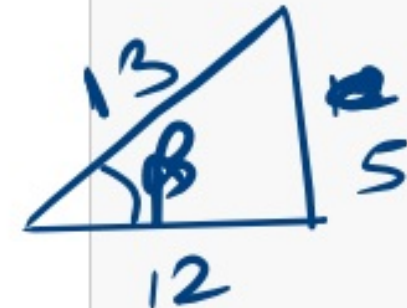
$\cos^{-1}\frac{4}{5} = \alpha$

$\Rightarrow \frac{4}{5} = \cos\alpha$



$\cos^{-1}\frac{12}{13} = \beta$

$\Rightarrow \frac{12}{13} = \cos\beta$



$\sin\beta = \frac{5}{13}$

$\cos(\alpha + \beta) = \cos\alpha\cos\beta - \sin\alpha\sin\beta$

$\cos\left(\cos^{-1}\frac{4}{5} + \cos^{-1}\frac{12}{13}\right) = \frac{4}{5} \times \frac{12}{13} - \frac{3}{5} \times \frac{5}{13}$

$\Rightarrow \cos\left(\cos^{-1}\frac{4}{5} + \cos^{-1}\frac{12}{13}\right) = \frac{48}{65} - \frac{15}{65} = \frac{33}{65}$

H.W

Exercise 2.2 - Q8, Q10 to Q15

Example-6

Misc Exercise - Q1 to Q7

8. $\tan^{-1} \sqrt{x} = \frac{1}{2} \cos^{-1} \frac{1-x}{1+x}, x \in [0, 1]$

Sol

RHS

$$\frac{1}{2} \cos^{-1} \left(\frac{1-x}{1+x} \right)$$

Let $x = \tan^2 \theta$

$$\Rightarrow \tan \theta = \sqrt{x}$$

$$\Rightarrow \underline{\theta = \tan^{-1}(\sqrt{x})}$$

$$\frac{1}{2} \cos^{-1} \left(\frac{1 - \tan^2 \theta}{1 + \tan^2 \theta} \right)$$

$$\frac{1}{2} \cos^{-1} (\cos 2\theta)$$

$$\frac{1}{2} (2\theta)$$

θ

$$\tan^{-1} \sqrt{x} = \text{L.H.S.}$$

$$\text{R.H.S.} = \text{L.H.S.}$$

Hence proved

$$\frac{1-x}{1+x}$$

Put $x = \tan^2 \theta$

10. $\tan^{-1} \left(\frac{\sqrt{1+x} - \sqrt{1-x}}{\sqrt{1+x} + \sqrt{1-x}} \right) = \frac{\pi}{4} - \frac{1}{2} \cos^{-1} x, -\frac{1}{\sqrt{2}} \leq x \leq 1$ [Hint: Put $x = \cos 2\theta$]

V. 24

Thumb rule

$$\frac{\sqrt{1+x} - \sqrt{1-x}}{\sqrt{1+x} + \sqrt{1-x}}$$

L.H.S

$$\tan^{-1} \left(\frac{\sqrt{1+\cos 2\theta} - \sqrt{1-\cos 2\theta}}{\sqrt{1+\cos 2\theta} + \sqrt{1-\cos 2\theta}} \right)$$

$$\tan^{-1} \left(\frac{\sqrt{2\cos^2\theta} - \sqrt{2\sin^2\theta}}{\sqrt{2\cos^2\theta} + \sqrt{2\sin^2\theta}} \right)$$

$$\tan^{-1} \left(\frac{\sqrt{2} (\cos\theta - \sin\theta)}{\sqrt{2} (\cos\theta + \sin\theta)} \right)$$

put $x = \cos 2\theta$

$$x = \cos 2\theta$$

$$\Rightarrow \cos^{-1} x = 2\theta$$

$$\Rightarrow \theta = \frac{1}{2} \cos^{-1} x$$

Rough

$$1 + \cos 2\theta = 2\cos^2\theta$$

$$1 - \cos 2\theta = 2\sin^2\theta$$

$$\tan^{-1} \left(\frac{\cos \theta - \sin \theta}{\cos \theta + \sin \theta} \right)$$

Divide by $\cos \theta$ in N° , D° in the expression $\frac{\cos \theta - \sin \theta}{\cos \theta + \sin \theta}$

$$\tan^{-1} \left(\frac{1 - \tan \theta}{1 + \tan \theta} \right) \rightarrow 1. \tan \theta$$

$$\tan^{-1} \left(\frac{\tan \frac{\pi}{4} - \tan \theta}{1 + \tan \frac{\pi}{4} \tan \theta} \right)$$

$$\tan^{-1} \left(\tan \left(\frac{\pi}{4} - \theta \right) \right)$$

$$\Rightarrow \frac{\pi}{4} - \theta$$

$$\Rightarrow \frac{\pi}{4} - \frac{1}{2} \cos^{-1} x$$

$$R.H.S = L.H.S$$

Hence proved

9. $\cot^{-1} \left(\frac{\sqrt{1+\sin x} + \sqrt{1-\sin x}}{\sqrt{1+\sin x} - \sqrt{1-\sin x}} \right) = \frac{x}{2}, x \in \left(0, \frac{\pi}{4} \right)$ V. V. V.

sol

$$\cot^{-1} \left(\frac{\sqrt{1 + \cos\left(\frac{\pi}{2} - x\right)} + \sqrt{1 - \cos\left(\frac{\pi}{2} - x\right)}}{\sqrt{1 + \cos\left(\frac{\pi}{2} - x\right)} - \sqrt{1 - \cos\left(\frac{\pi}{2} - x\right)}} \right)$$

$$\cot^{-1} \left(\frac{\sqrt{2 \cos^2\left(\frac{\pi}{4} - \frac{x}{2}\right)} + \sqrt{2 \sin^2\left(\frac{\pi}{4} - \frac{x}{2}\right)}}{\sqrt{2 \cos^2\left(\frac{\pi}{4} - \frac{x}{2}\right)} - \sqrt{2 \sin^2\left(\frac{\pi}{4} - \frac{x}{2}\right)}} \right)$$

$$\cot^{-1} \left(\frac{\sqrt{2} \left(\cos\left(\frac{\pi}{4} - \frac{x}{2}\right) + \sin\left(\frac{\pi}{4} - \frac{x}{2}\right) \right)}{\sqrt{2} \left(\cos\left(\frac{\pi}{4} - \frac{x}{2}\right) - \sin\left(\frac{\pi}{4} - \frac{x}{2}\right) \right)} \right)$$

$$1 + \cos 2\theta = 2 \cos^2 \theta$$

$$\cot^{-1} \left(\frac{\sqrt{2} \left(\cos \left(\frac{\pi}{4} - \frac{x}{2} \right) + \sin \left(\frac{\pi}{4} - \frac{x}{2} \right) \right)}{\sqrt{2} \left(\cos \left(\frac{\pi}{4} - \frac{x}{2} \right) - \sin \left(\frac{\pi}{4} - \frac{x}{2} \right) \right)} \right)$$

⇒ Divide by $\cos \left(\frac{\pi}{4} - \frac{x}{2} \right)$ in the expression

$$\Rightarrow \cot^{-1} \left(\frac{1 + \tan \left(\frac{\pi}{4} - \frac{x}{2} \right)}{1 - \tan \left(\frac{\pi}{4} - \frac{x}{2} \right)} \right)$$

$$\Rightarrow \cot^{-1} \left(\frac{\tan \frac{\pi}{4} + \tan \left(\frac{\pi}{4} - \frac{x}{2} \right)}{1 - \tan \frac{\pi}{4} \tan \left(\frac{\pi}{4} - \frac{x}{2} \right)} \right)$$

$$\cot^{-1} \left(\tan \left(\frac{\pi}{4} + \frac{\pi}{4} - \frac{x}{2} \right) \right)$$

$$\cot^{-1} \left(\tan \left(\frac{\pi}{2} - \frac{x}{2} \right) \right)$$

$$\cot^{-1} \left(\cot \frac{x}{2} \right)$$

$$\frac{x}{2}$$

$$\text{LHS} = \text{RHS}$$

hence proved

Prove that: $\tan^{-1} \left\{ \frac{a \cos x - b \sin x}{b \cos x + a \sin x} \right\} = \tan^{-1} \left(\frac{a}{b} \right) - x$

V. BH

Sol L.H.S

$$\tan^{-1} \left(\frac{a \cos x - b \sin x}{b \cos x + a \sin x} \right)$$

Divide by $b \cos x$ in Nr, or of this expren, we get

$$\tan^{-1} \left(\frac{\frac{a}{b} - \tan x}{1 + \frac{a}{b} \tan x} \right)$$

$$\text{Let } \frac{a}{b} = \tan \theta \Rightarrow \theta = \tan^{-1} \left(\frac{a}{b} \right)$$

$$\tan^{-1} \left(\frac{\tan \theta - \tan x}{1 + \tan \theta \tan x} \right)$$

$$\tan^{-1} (\tan (\theta - x))$$

$$\theta - x$$
$$\tan^{-1} \left(\frac{a}{b} \right) - x$$

$$\tan (c + d)$$

$$= \frac{\tan c + \tan d}{1 + \tan c \tan d}$$

Simplify: $\sin^{-1} \left\{ \frac{\sin x + \cos x}{\sqrt{2}} \right\}, -\frac{\pi}{4} < x < \frac{\pi}{4}$

$$\sin^{-1} \left(\frac{1}{\sqrt{2}} \sin x + \frac{1}{\sqrt{2}} \cos x \right)$$

$$\sin^{-1} \left(\sin x \cos \frac{\pi}{4} + \cos x \sin \frac{\pi}{4} \right)$$

$$\sin^{-1} \left(\sin \left(x + \frac{\pi}{4} \right) \right)$$

$$x + \frac{\pi}{4}$$

✓

$$\sin(A+B)$$

$$= \sin A \cos B + \cos A \sin B$$

$$\tan^{-1}\left(\frac{\sqrt{1+x^2} + \sqrt{1-x^2}}{\sqrt{1+x^2} - \sqrt{1-x^2}}\right) = \frac{\pi}{4} + \frac{1}{2} \cos^{-1} x^2$$

Proof

$$\text{Let } x^2 = \cos 2\theta$$

$$\Rightarrow 2\theta = \cos^{-1}(x^2)$$

$$\Rightarrow \theta = \frac{1}{2} \cos^{-1}(x^2)$$

$$\tan^{-1}\left(\frac{\sqrt{1+\cos 2\theta} + \sqrt{1-\cos 2\theta}}{\sqrt{1+\cos 2\theta} - \sqrt{1-\cos 2\theta}}\right)$$

intermediate steps

$$\tan^{-1}\left(\tan\left(\frac{\pi}{4} + \theta\right)\right)$$

$$\frac{\pi}{4} + \theta$$

$$\frac{\pi}{4} + \frac{1}{2} \cos^{-1}(x^2)$$

$$\text{LHS} = \text{RHS}$$

Hence proved

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Q1

$$2 \tan^{-1}(\cos x) = \tan^{-1}(2 \operatorname{cosec} x) = 0 \text{ (let)}$$

$$2 \tan^{-1}(\cos x) = 0$$

$$\tan^{-1}(\cos x) = \frac{0}{2}$$

$$\Rightarrow \boxed{\cos x = \tan \frac{0}{2}}$$

$$\tan^{-1}(2 \operatorname{cosec} x) = 0$$

$$\boxed{2 \operatorname{cosec} x = \tan 0}$$

$$7. \tan^{-1} \frac{1}{16} = \sin^{-1} \frac{1}{13} + \cos^{-1} \frac{1}{5}$$

Prove that

$$8. \tan^{-1} \sqrt{x} = \frac{1}{2} \cos^{-1} \frac{1-x}{1+x}, x \in [0, 1]$$

$$9. \cot^{-1} \left(\frac{\sqrt{1+\sin x} + \sqrt{1-\sin x}}{\sqrt{1+\sin x} - \sqrt{1-\sin x}} \right) = \frac{x}{2}, x \in \left(0, \frac{\pi}{4}\right)$$

$$10. \tan^{-1} \left(\frac{\sqrt{1+x} - \sqrt{1-x}}{\sqrt{1+x} + \sqrt{1-x}} \right) = \frac{\pi}{4} - \frac{1}{2} \cos^{-1} x, -\frac{1}{\sqrt{2}} \leq x \leq 1 \text{ [Hint: Put } x = \cos 2\theta]$$

Solve the following equations:

$$11. 2 \tan^{-1}(\cos x) = \tan^{-1}(2 \operatorname{cosec} x) \quad 12. \tan^{-1} \frac{1-x}{1+x} = \frac{1}{2} \tan^{-1} x, (x > 0)$$

$$13. \sin(\tan^{-1} x), |x| < 1 \text{ is equal to}$$

$$(A) \frac{x}{\sqrt{1-x^2}} \quad (B) \frac{1}{\sqrt{1-x^2}} \quad (C) \frac{1}{\sqrt{1+x^2}} \quad (D) \frac{x}{\sqrt{1+x^2}}$$

$$14. \sin^{-1}(1-x) - 2 \sin^{-1} x = \frac{\pi}{2}, \text{ then } x \text{ is equal to}$$

$$(A) 0, \frac{1}{2} \quad (B) 1, \frac{1}{2} \quad (C) 0 \quad (D) \frac{1}{2}$$

$$\Rightarrow \sin x = \cos x$$

$$\Rightarrow \tan x = 1 \Rightarrow x = \frac{\pi}{4}$$

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$$\boxed{\tan 2A = \frac{2 \tan A}{1 - \tan^2 A}}$$

$$\tan 0 = \frac{2 \tan 0/2}{1 - \tan^2 0/2}$$

$$2 \operatorname{cosec} x = \frac{2 \cos x}{1 - \cos^2 x}$$

$$\operatorname{cosec} x = \frac{\cos x}{\sin^2 x}$$

$$\operatorname{cosec} x \cdot \sin^2 x = \cos x$$

$$\frac{1}{\sin x} \cdot \sin^2 x = \cos x$$



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12. $\tan^{-1} \frac{1-x}{1+x} = \frac{1}{2} \tan^{-1} x, (x > 0)$

84 $\Rightarrow \tan^{-1} \frac{1-x}{1+x} = \frac{1}{2} \tan^{-1} x = \theta \text{ (let)}$

$\tan^{-1} \frac{1-x}{1+x} = \theta$

$\Rightarrow \left[\frac{1-x}{1+x} = \tan \theta \right]$

$\frac{1}{2} \tan^{-1} x = \theta$

$\Rightarrow \tan^{-1} x = 2\theta$

$\Rightarrow \boxed{x = \tan 2\theta}$

$x = \pm \frac{1}{\sqrt{3}}$
 $x = \frac{1}{\sqrt{3}}; x > 0$

$\tan 2\theta = \frac{2 \tan \theta}{1 - \tan^2 \theta}$

$x = \frac{2 \left(\frac{1-x}{1+x} \right)}{1 - \left(\frac{1-x}{1+x} \right)^2}$

$\Rightarrow x = \frac{2 \left(\frac{1-x}{1+x} \right)}{\frac{(1+x)^2 - (1-x)^2}{(1+x)^2}}$

$\Rightarrow x = \frac{2(1-x)(1+x)}{(1+x^2+2x) - (1+x^2-2x)}$

$\Rightarrow x = \frac{2x(1-x^2)}{2 \cancel{4} x}$

$\Rightarrow 2x^2 = 1 - x^2$

$\Rightarrow 3x^2 = 1$



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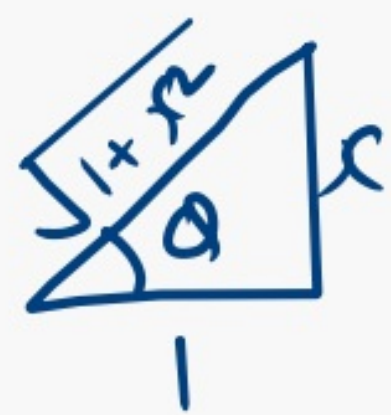
Q13

$\sin(\tan^{-1} x) = ?$

$\tan^{-1} x = \theta$ (let)

$\Rightarrow \boxed{\tan \theta = x}$

$\sin \theta = ?$



$\sin \theta = \frac{x}{\sqrt{1+x^2}}$

8. $\tan^{-1} \sqrt{x} = \frac{1}{2} \cos^{-1} \frac{1-x}{1+x}, x \in [0, 1]$
9. $\cot^{-1} \left(\frac{\sqrt{1+\sin x} + \sqrt{1-\sin x}}{\sqrt{1+\sin x} - \sqrt{1-\sin x}} \right) = \frac{x}{2}, x \in \left(0, \frac{\pi}{4}\right)$
10. $\tan^{-1} \left(\frac{\sqrt{1+x} - \sqrt{1-x}}{\sqrt{1+x} + \sqrt{1-x}} \right) = \frac{\pi}{4} - \frac{1}{2} \cos^{-1} x, -\frac{1}{\sqrt{2}} \leq x \leq 1$ [Hint: Put $x = \cos 2\theta$]

Solve the following equations:

11. $2 \tan^{-1}(\cos x) = \tan^{-1}(2 \operatorname{cosec} x)$
12. $\tan^{-1} \frac{1-x}{1+x} = \frac{1}{2} \tan^{-1} x, (x > 0)$
13. $\sin(\tan^{-1} x), |x| < 1$ is equal to
 (A) $\frac{x}{\sqrt{1-x^2}}$ (B) $\frac{1}{\sqrt{1-x^2}}$ (C) $\frac{1}{\sqrt{1+x^2}}$ (D) $\frac{x}{\sqrt{1+x^2}}$
14. $\sin^{-1}(1-x) - 2 \sin^{-1} x = \frac{\pi}{2}$, then x is equal to
 (A) $0, \frac{1}{2}$ (B) $1, \frac{1}{2}$ (C) 0 (D) $\frac{1}{2}$

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$\cos(\tan^{-1} x)$

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Q14

$$\sin^{-1}(1-x) = \frac{\pi}{2} + 2\sin^{-1}x$$

$$\Rightarrow 1-x = \sin\left(\frac{\pi}{2} + 2\sin^{-1}x\right)$$

$$\Rightarrow 1-x = \cos(2\sin^{-1}x)$$

$$\Rightarrow 1-x = \cos(2\sin^{-1}x)$$

$$\text{Let } 2\sin^{-1}x = \theta$$

$$\sin^{-1}x = \frac{\theta}{2}$$

$$(x = \sin \frac{\theta}{2})$$

$$8. \tan^{-1}\sqrt{x} = \frac{1}{2}\cos^{-1}\frac{1-x}{1+x}, x \in [0, 1]$$

$$9. \cot^{-1}\left(\frac{\sqrt{1+\sin x} + \sqrt{1-\sin x}}{\sqrt{1+\sin x} - \sqrt{1-\sin x}}\right) = \frac{x}{2}, x \in \left(0, \frac{\pi}{4}\right)$$

$$10. \tan^{-1}\left(\frac{\sqrt{1+x} - \sqrt{1-x}}{\sqrt{1+x} + \sqrt{1-x}}\right) = \frac{\pi}{4} - \frac{1}{2}\cos^{-1}x, -\frac{1}{\sqrt{2}} \leq x \leq 1 \text{ [Hint: Put } x = \cos 2\theta]$$

Solve the following equations:

$$11. 2\tan^{-1}(\cos x) = \tan^{-1}(2 \operatorname{cosec} x) \quad 12. \tan^{-1}\frac{1-x}{1+x} = \frac{1}{2}\tan^{-1}x, (x > 0)$$

$$13. \sin(\tan^{-1}x), |x| < 1 \text{ is equal to}$$

$$(A) \frac{x}{\sqrt{1-x^2}} \quad (B) \frac{1}{\sqrt{1-x^2}} \quad (C) \frac{1}{\sqrt{1+x^2}} \quad (D) \frac{x}{\sqrt{1+x^2}}$$

$$14. \sin^{-1}(1-x) - 2\sin^{-1}x = \frac{\pi}{2}, \text{ then } x \text{ is equal to}$$

$$(A) 0, \frac{1}{2} \quad (B) 1, \frac{1}{2} \quad (C) 0 \quad (D) \frac{1}{2}$$

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$$\cos \theta = 1 - 2\sin^2 \frac{\theta}{2}$$

$$\cos \theta = 1 - 2x^2$$

$$1-x = 1 - 2x^2$$

$$-x = -2x^2$$

$$2x^2 - x = 0$$

$$x(2x - 1) = 0$$

$$x = 0 \text{ or } x = \frac{1}{2}$$

H.W

Mus Exercise :- 08 to 014

+
Extra questions that we did in today's class

5. Prove that:

$$\tan^{-1}\left(\frac{\sqrt{1+x^2} + \sqrt{1-x^2}}{\sqrt{1+x^2} - \sqrt{1-x^2}}\right) = \frac{\pi}{4} + \frac{1}{2} \cos^{-1} x^2$$

Simplify :

$$\sin^{-1}\left\{\frac{\sin x + \cos x}{\sqrt{2}}\right\}, -\frac{\pi}{4} < x < \frac{\pi}{4}$$

6.

Prove that : $\tan^{-1}\left\{\frac{a \cos x - b \sin x}{b \cos x + a \sin x}\right\} = \tan^{-1}\left(\frac{a}{b}\right) - x$

Integration (anti-derivative)

It is nothing but reverse of differentiation.

$$\frac{d}{dx}(\sin x) = \cos x$$

$$\frac{d}{dx}(\cos x) = -\sin x$$

$$\frac{d}{dx}(\tan x) = \sec^2 x$$

$$\frac{d}{dx}(\text{Constant}) = 0$$

$$\int \cos x \, dx = \sin x$$

$$\int \sin x \, dx = -\cos x$$

$$\int \sec^2 x \, dx = \tan x$$

$$\int 0 \cdot dx = \text{Constant } (c)$$

$$\frac{d}{dx}(\sin x + C) = \cos x$$

$$\int \cos x \, dx = \sin x + C$$

$$\frac{d}{dx}(\cot x + C) = -\operatorname{cosec}^2 x$$

$$\int \operatorname{cosec}^2 x \, dx = -\cot x + C$$

In general,

$$\frac{d}{dx}(f(x)) = f'(x)$$

$\Rightarrow$

$$\int f'(x) \, dx = f(x) + C$$

other name of integration is anti-derivative.

Annotations toolbar with icons for drawing, erasing, and text tools.

$\frac{d}{dx} f(x)$
↓
derivative of f
w.r.t x

class of anti derivatives read as the indefinite integral of f with respect to x .

Symbolically, we write $\int f(x) dx = F(x) + C$.

Notation Given that $\frac{dy}{dx} = f(x)$, we write $y = \int f(x) dx$.

For the sake of convenience, we mention below the following symbols/terms/phrases with their meanings as given in the Table (7.1).

Table 7.1

Symbols/Terms/Phrases	Meaning
$\int f(x) dx$ ✓	Integral of f with respect to x
<u>$f(x)$ in $\int f(x) dx$</u>	<u>Integrand</u>
<u>x in $\int f(x) dx$</u>	<u>Variable of integration</u>
Integrate	Find the integral
An integral of f	A function F such that $F'(x) = f(x)$
Integration	The process of finding the integral
<u>Constant of Integration</u>	Any real number <u>C</u> , considered as constant function



formulas

(i) $\frac{1}{(n+1)} \cdot (n+1)x^n$

$$\int \frac{1}{\sqrt{x}} dx = \int x^{-1/2} dx$$

$$= \frac{x^{-1/2+1}}{-1/2+1}$$

$$= \frac{x^{1/2}}{1/2} + C$$

$$\int \frac{1}{x^{8/9}} dx = \int x^{-8/9} dx$$

$$= \frac{x^{-8/9+1}}{-8/9+1} + C$$

Derivatives

(i) $\frac{d}{dx} \left(\frac{x^{n+1}}{n+1} \right) = x^n$;

Particularly, we note that

$\frac{d}{dx}(x) = 1$;

(ii) $\frac{d}{dx}(\sin x) = \cos x$;

(iii) $\frac{d}{dx}(-\cos x) = \sin x$;

(iv) $\frac{d}{dx}(\tan x) = \sec^2 x$;

(v) $\frac{d}{dx}(-\cot x) = \operatorname{cosec}^2 x$;

(vi) $\frac{d}{dx}(\sec x) = \sec x \tan x$;

(vii) $\frac{d}{dx}(-\operatorname{cosec} x) = \operatorname{cosec} x \cot x$;

(viii) $\frac{d}{dx}(\sin^{-1} x) = \frac{1}{\sqrt{1-x^2}}$;

(ix) $\frac{d}{dx}(-\cos^{-1} x) = \frac{1}{\sqrt{1-x^2}}$;

(x) $\frac{d}{dx}(\tan^{-1} x) = \frac{1}{1+x^2}$.

Integrals (Anti derivatives)

$$\int x^n dx = \frac{x^{n+1}}{n+1} + C, n \neq -1$$

$$\int dx = x + C$$

$$\int \cos x dx = \sin x + C$$

$$\int \sin x dx = -\cos x + C$$

$$\int \sec^2 x dx = \tan x + C$$

$$\int \operatorname{cosec}^2 x dx = -\cot x + C$$

$$\int \sec x \tan x dx = \sec x + C$$

$$\int \operatorname{cosec} x \cot x dx = -\operatorname{cosec} x + C$$

$$\int \frac{dx}{\sqrt{1-x^2}} = \sin^{-1} x + C$$

$$\int \frac{dx}{\sqrt{1-x^2}} = -\cos^{-1} x + C$$

$$\int \frac{dx}{1+x^2} = \tan^{-1} x + C$$

eg:- $\int x^2 dx = \frac{x^{2+1}}{2+1}$

$$= \frac{x^3}{3} + C$$

$$\int x^4 dx = \frac{x^5}{5} + C$$

$$\int \sqrt{x} dx = \frac{x^{3/2}}{3/2} + C$$

$$\int x^{3/8} dx = \frac{x^{11/8}}{11/8} + C$$

$$\int x^{-1} dx = \frac{x^{-1+1}}{-1+1} + C$$

↑
Zero

$$\int \frac{1}{x} dx = \log x + C$$

↓
 $\log e^x$

Derivatives

- (i) $\frac{d}{dx} \left(\frac{x^{n+1}}{n+1} \right) = x^n$;
Particularly, we note that
 $\frac{d}{dx}(x) = 1$;
- (ii) $\frac{d}{dx}(\sin x) = \cos x$;
- (iii) $\frac{d}{dx}(-\cos x) = \sin x$;
- (iv) $\frac{d}{dx}(\tan x) = \sec^2 x$;
- (v) $\frac{d}{dx}(-\cot x) = \operatorname{cosec}^2 x$;
- (vi) $\frac{d}{dx}(\sec x) = \sec x \tan x$;
- (vii) $\frac{d}{dx}(-\operatorname{cosec} x) = \operatorname{cosec} x \cot x$;
- (viii) $\frac{d}{dx}(\sin^{-1} x) = \frac{1}{\sqrt{1-x^2}}$;
- (ix) $\frac{d}{dx}(-\cos^{-1} x) = \frac{1}{\sqrt{1-x^2}}$;
- (x) $\frac{d}{dx}(\tan^{-1} x) = \frac{1}{1+x^2}$.

Integrals (Anti derivatives)

- $$\int x^n dx = \frac{x^{n+1}}{n+1} + C, n \neq -1$$
- $$\int dx = x + C$$
- $$\int \cos x dx = \sin x + C$$
- $$\int \sin x dx = -\cos x + C$$
- $$\int \sec^2 x dx = \tan x + C$$
- $$\int \operatorname{cosec}^2 x dx = -\cot x + C$$
- $$\int \sec x \tan x dx = \sec x + C$$
- $$\int \operatorname{cosec} x \cot x dx = -\operatorname{cosec} x + C$$
- $$\int \frac{dx}{\sqrt{1-x^2}} = \sin^{-1} x + C$$
- $$\int \frac{dx}{\sqrt{1-x^2}} = -\cos^{-1} x + C$$
- $$\int \frac{dx}{1+x^2} = \tan^{-1} x + C$$

$$\frac{d}{dx}(x) = 1$$

$$\int 1 dx = x + C$$

$$\int dx = x + C$$

$$\int 2 dx = 2 \int dx = 2x + C$$

Derivatives

$$(i) \frac{d}{dx} \left(\frac{x^{n+1}}{n+1} \right) = x^n ;$$

Particularly, we note that

$$\frac{d}{dx}(x) = 1 ;$$

$$(ii) \frac{d}{dx}(\sin x) = \cos x ;$$

$$(iii) \frac{d}{dx}(-\cos x) = \sin x ;$$

$$(iv) \frac{d}{dx}(\tan x) = \sec^2 x ;$$

$$(v) \frac{d}{dx}(-\cot x) = \operatorname{cosec}^2 x ;$$

$$(vi) \frac{d}{dx}(\sec x) = \sec x \tan x ;$$

$$(vii) \frac{d}{dx}(-\operatorname{cosec} x) = \operatorname{cosec} x \cot x ;$$

$$(viii) \frac{d}{dx}(\sin^{-1} x) = \frac{1}{\sqrt{1-x^2}} ;$$

$$(ix) \frac{d}{dx}(-\cos^{-1} x) = \frac{1}{\sqrt{1-x^2}} ;$$

$$(x) \frac{d}{dx}(\tan^{-1} x) = \frac{1}{1+x^2} .$$

Integrals (Anti derivatives)

$$\int x^n dx = \frac{x^{n+1}}{n+1} + C, n \neq -1$$

$$\int dx = x + C$$

$$\int \cos x dx = \sin x + C$$

$$\int \sin x dx = -\cos x + C$$

$$\int \sec^2 x dx = \tan x + C$$

$$\int \operatorname{cosec}^2 x dx = -\cot x + C$$

$$\int \sec x \tan x dx = \sec x + C$$

$$\int \operatorname{cosec} x \cot x dx = -\operatorname{cosec} x + C$$

$$\int \frac{dx}{\sqrt{1-x^2}} = \sin^{-1} x + C$$

$$\int \frac{dx}{\sqrt{1-x^2}} = -\cos^{-1} x + C$$

$$\int \frac{dx}{1+x^2} = \tan^{-1} x + C$$

Derivatives

$$(i) \frac{d}{dx} \left(\frac{x^{n+1}}{n+1} \right) = x^n ;$$

Particularly, we note that

$$\frac{d}{dx}(x) = 1 ;$$

$$(ii) \frac{d}{dx}(\sin x) = \cos x ;$$

$$(iii) \frac{d}{dx}(-\cos x) = \sin x ;$$

$$(iv) \frac{d}{dx}(\tan x) = \sec^2 x ;$$

$$(v) \frac{d}{dx}(-\cot x) = \operatorname{cosec}^2 x ;$$

$$(vi) \frac{d}{dx}(\sec x) = \sec x \tan x ;$$

$$(vii) \frac{d}{dx}(-\operatorname{cosec} x) = \operatorname{cosec} x \cot x ;$$

$$(viii) \frac{d}{dx}(\sin^{-1} x) = \frac{1}{\sqrt{1-x^2}} ;$$

$$(ix) \frac{d}{dx}(-\cos^{-1} x) = \frac{1}{\sqrt{1-x^2}} ;$$

$$(x) \frac{d}{dx}(\tan^{-1} x) = \frac{1}{1+x^2} .$$

Integrals (Anti derivatives)

$$\int x^n dx = \frac{x^{n+1}}{n+1} + C, n \neq -1$$

$$\int dx = x + C$$

$$\int \cos x dx = \sin x + C$$

$$\int \sin x dx = -\cos x + C$$

$$\int \sec^2 x dx = \tan x + C$$

$$\int \operatorname{cosec}^2 x dx = -\cot x + C$$

$$\int \sec x \tan x dx = \sec x + C$$

$$\int \operatorname{cosec} x \cot x dx = -\operatorname{cosec} x + C$$

$$\int \frac{dx}{\sqrt{1-x^2}} = \sin^{-1} x + C$$

$$\int \frac{dx}{\sqrt{1-x^2}} = -\cos^{-1} x + C$$

$$\int \frac{dx}{1+x^2} = \tan^{-1} x + C$$

formula of integration
of Trigonometric
functions.

$$\frac{d}{dx}(\cos x) = -\sin x$$

$$\frac{d}{dx}(\cot x) = -\operatorname{cosec}^2 x$$

$$\frac{d}{dx}(\operatorname{cosec} x) = -\operatorname{cosec} x \cot x$$



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$$\frac{d}{dx}(\cos^{-1}x) = \frac{-1}{\sqrt{1-x^2}}$$

$$\frac{d}{dx}(\cot^{-1}x) = \frac{-1}{1+x^2} + C$$

$$\frac{d}{dx}(\sec^{-1}x) = \frac{1}{|x|\sqrt{x^2-1}}$$

$$\frac{d}{dx}(\operatorname{cosec}^{-1}x) = \frac{-1}{|x|\sqrt{x^2-1}}$$

- (v) $\frac{d}{dx}(-\cot x) = \operatorname{cosec}^2 x$; $\int \operatorname{cosec}^2 x \, dx = -\cot x + C$
- (vi) $\frac{d}{dx}(\sec x) = \sec x \tan x$; $\int \sec x \tan x \, dx = \sec x + C$
- (vii) $\frac{d}{dx}(-\operatorname{cosec} x) = \operatorname{cosec} x \cot x$; $\int \operatorname{cosec} x \cot x \, dx = -\operatorname{cosec} x + C$
- (viii) $\frac{d}{dx}(\sin^{-1} x) = \frac{1}{\sqrt{1-x^2}}$; $\int \frac{dx}{\sqrt{1-x^2}} = \sin^{-1} x + C$
- (ix) $\frac{d}{dx}(-\cos^{-1} x) = \frac{1}{\sqrt{1-x^2}}$; $\int \frac{dx}{\sqrt{1-x^2}} = -\cos^{-1} x + C$
- (x) $\frac{d}{dx}(\tan^{-1} x) = \frac{1}{1+x^2}$; $\int \frac{dx}{1+x^2} = \tan^{-1} x + C$
- (xi) $\frac{d}{dx}(e^x) = e^x$; $\int e^x \, dx = e^x + C$

Formula of integration of ITF

$$\int \frac{dx}{\sqrt{1-x^2}} = \sin^{-1} x + C = -\cos^{-1} x + C$$

$$\int \frac{dx}{\sqrt{1-x^2}} = -\cos^{-1} x + C$$

$$\int \frac{dx}{1+x^2} = \tan^{-1} x + C = -\cot^{-1} x + C$$

$$\int e^x \, dx = e^x + C$$

$$\int \frac{dx}{|x|\sqrt{x^2-1}} = \sec^{-1} x + C = -\operatorname{cosec}^{-1} x + C$$

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$$\int \frac{1}{x} \, dx = \log |x| + C$$



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$$\frac{d}{dx}(a^x) = a^x \log a$$

$$\int a^x \log a \, dx = a^x + C$$

$$\int a^x \, dx = \frac{a^x}{\log a}$$

$$\frac{d}{dx}(\log x) = \frac{1}{x}$$

$$\int \frac{1}{x} \, dx = \log |x| + C$$

$$(x) \frac{d}{dx} \left(\frac{1}{1+x^2} \right) = \frac{-2x}{(1+x^2)^2}$$

$$(xi) \frac{d}{dx}(e^x) = e^x ;$$

$$\int e^x \, dx = e^x + C$$

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$$(xii) \frac{d}{dx}(\log |x|) = \frac{1}{x} ;$$

$$\int \frac{1}{x} \, dx = \log |x| + C$$

$$(xiii) \frac{d}{dx} \left(\frac{a^x}{\log a} \right) = a^x ;$$

$$\int a^x \, dx = \frac{a^x}{\log a} + C$$

Note In practice, we normally do not mention the interval over which the various functions are defined. However, in any specific problem one has to keep it in mind.

7.2.1 Some properties of indefinite integral

In this sub section, we shall derive some properties of indefinite integrals.

- (I) The process of differentiation and integration are inverses of each other in the sense of the following results :

$$\frac{d}{dx} \int f(x) \, dx = f(x)$$

and

$$\int f'(x) \, dx = f(x) + C, \text{ where } C \text{ is any arbitrary constant.}$$

Properties of Integration

① $\frac{d}{dx} f(x) = f'(x)$

$\int f(x) dx = F(x) + C$

②

$\int (f(x) \pm g(x)) dx = \int f(x) dx \pm \int g(x) dx$

③ $\int k f(x) dx = k \int f(x) dx$

eg $\int 2 \sin x dx = 2 \int \sin x dx$

$= -2 \cos x + C$

④ $\int (k_1 f_1(x) + k_2 f_2(x) + \dots) dx$

$= k_1 \int f_1(x) dx + k_2 \int f_2(x) dx + \dots$

Proof By the Property (I), $\frac{d}{dx} \int k f(x) dx = k f(x)$.

Also $\frac{d}{dx} \left[k \int f(x) dx \right] = k \frac{d}{dx} \int f(x) dx = k f(x)$

Therefore, using the Property (II), we have $\int k f(x) dx = k \int f(x) dx$.

(V) Properties (III) and (IV) can be generalised to a finite number of functions $f_1, f_2, \dots, f_n$ and the real numbers, $k_1, k_2, \dots, k_n$ giving

$$\begin{aligned} & \int [k_1 f_1(x) + k_2 f_2(x) + \dots + k_n f_n(x)] dx \\ &= k_1 \int f_1(x) dx + k_2 \int f_2(x) dx + \dots + k_n \int f_n(x) dx. \end{aligned}$$

To find an anti derivative of a given function, we search intuitively for a function whose derivative is the given function. The search for the requisite function for finding an anti derivative is known as integration by the method of inspection. We illustrate it through some examples.

Example 1 Write an anti derivative for each of the following functions using the method of inspection:

(i) $\cos 2x$

(ii) $3x^2 + 4x^3$

(iii) $\frac{1}{x}, x \neq 0$

Solution

(i) We look for a function whose derivative is $\cos 2x$. Recall that



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Eg 1 (ii)

$$\int (3x^2 + 4x^3) dx$$

$$= 3 \int x^2 dx + 4 \int x^3 dx$$

$$= 3 \cdot \frac{x^3}{3} + 4 \cdot \frac{x^4}{4} + C$$

$$= x^3 + x^4 + C$$

(iii) we know that

$$\frac{d}{dx}(\log x) = \frac{1}{x}, x > 0 \text{ and } \frac{d}{dx}[\log(-x)] = \frac{1}{-x}(-1) = \frac{1}{x}, x < 0$$

$$\text{Combining above, we get } \frac{d}{dx}(\log|x|) = \frac{1}{x}, x \neq 0$$

$$\text{Therefore, } \int \frac{1}{x} dx = \log|x| \text{ is one of the anti derivatives of } \frac{1}{x}$$

Example 2 Find the following integrals:

(i) $\int \frac{x^3 - 1}{x^2} dx$

(ii) $\int (x^{\frac{2}{3}} + 1) dx$

(iii) $\int (x^{\frac{3}{2}} + 2e^x - \frac{1}{x}) dx$

Solution

(i) We have

$$\int \frac{x^3 - 1}{x^2} dx = \int x dx - \int x^{-2} dx \quad (\text{by Property V})$$

$$= \left(\frac{x^{1+1}}{1+1} + C_1 \right) - \left(\frac{x^{-2+1}}{-2+1} + C_2 \right); C_1, C_2 \text{ are constants of integration}$$

$$= \frac{x^2}{2} + C_1 - \frac{x^{-1}}{-1} - C_2 = \frac{x^2}{2} + \frac{1}{x} + C_1 - C_2$$

$$= \frac{x^2}{2} + \frac{1}{x} + C, \text{ where } C = C_1 - C_2 \text{ is another constant of integration.}$$

Note From now onwards, we shall write only one constant of integration in the final answer.

(ii) We have

Eg 2 (ii)

$$\int (x^{2/3} + 1) dx$$

$$\int x^{2/3} dx + \int dx$$

$$\frac{x^{5/3}}{5/3} + x + C$$

$$\frac{3}{5} x^{5/3} + x + C$$



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Eg² (ii)

$$\int \frac{x^3 - 1}{x^2} dx$$

$$\int \left(\frac{x^3}{x^2} - \frac{1}{x^2} \right) dx$$

$$\int x dx - \int x^{-2} dx$$

$$\frac{x^2}{2} - \frac{x^{-1}}{-1} + C$$

$$\frac{x^2}{2} + \frac{1}{x} + C$$

(iii) we know that
 $\frac{d}{dx}(\log x) = \frac{1}{x}, x > 0$ and $\frac{d}{dx}[\log(-x)] = \frac{1}{-x}(-1) = \frac{1}{x}, x < 0$
 Combining above, we get $\frac{d}{dx}(\log|x|) = \frac{1}{x}, x \neq 0$
 Therefore, $\int \frac{1}{x} dx = \log|x|$ is one of the anti derivatives of $\frac{1}{x}$.

Example 2 Find the following integrals:

- (i) $\int \frac{x^3 - 1}{x^2} dx$ (ii) $\int (x^{\frac{2}{3}} + 1) dx$ (iii) $\int (x^{\frac{3}{2}} + 2e^x - \frac{1}{x}) dx$

Solution

(i) We have

$$\begin{aligned} \int \frac{x^3 - 1}{x^2} dx &= \int x dx - \int x^{-2} dx \quad (\text{by Property V}) \\ &= \left(\frac{x^{1+1}}{1+1} + C_1 \right) - \left(\frac{x^{-2+1}}{-2+1} + C_2 \right); C_1, C_2 \text{ are constants of integration} \\ &= \frac{x^2}{2} + C_1 - \frac{x^{-1}}{-1} - C_2 = \frac{x^2}{2} + \frac{1}{x} + C_1 - C_2 \\ &= \frac{x^2}{2} + \frac{1}{x} + C, \text{ where } C = C_1 - C_2 \text{ is another constant of integration.} \end{aligned}$$

Note From now onwards, we shall write only one constant of integration in the final answer.

(ii) We have

$$\int \frac{x^3 - x^2 + 2}{\sqrt{x}} dx$$

$$\int \left(\frac{x^3}{\sqrt{x}} - \frac{x^2}{\sqrt{x}} + \frac{2}{\sqrt{x}} \right) dx$$

$$\int x^{5/2} dx - \int x^{3/2} dx + \int 2x^{-1/2} dx$$

$$\frac{x^{7/2}}{7/2} - \frac{x^{5/2}}{5/2} + \frac{2x^{1/2}}{1/2} + C$$

$$\frac{2}{7} x^{7/2} - \frac{2}{5} x^{5/2} + 4x^{1/2} + C$$



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Eg 1

$$\frac{dF(x)}{dx} = f(x)$$

$$\Rightarrow \int f(x) dx = F(x) + C$$

$$\int (4x^3 - 6) dx = F(x) + C$$

$$4 \cdot \frac{x^4}{4} - 6x = F(x) + C$$

$$x^4 - 6x = F(x) + C$$

$$= -\cot x - \operatorname{cosec} x + C$$

(iii) We have

$$\begin{aligned} \int \frac{1 - \sin x}{\cos^2 x} dx &= \int \frac{1}{\cos^2 x} dx - \int \frac{\sin x}{\cos^2 x} dx \\ &= \int \sec^2 x dx - \int \tan x \sec x dx \\ &= \tan x - \sec x + C \end{aligned}$$

Example 4 Find the anti derivative F of f defined by $f(x) = 4x^3 - 6$, where $F(0) = 3$

Solution One anti derivative of $f(x)$ is $x^4 - 6x$ since

$$\frac{d}{dx}(x^4 - 6x) = 4x^3 - 6$$

Therefore, the anti derivative F is given by

$$F(x) = x^4 - 6x + C, \text{ where } C \text{ is constant.}$$

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Put $x = 0$

$$0^4 - 6(0) = F(0) + C$$

$$0 = 3 + C$$

$$C = -3$$

$$\Rightarrow F(x) = x^4 - 6x + 3$$

Given that

$$F(0) = 3, \text{ which gives,}$$

$$3 = 0 - 6 \times 0 + C \text{ or } C = 3$$

Hence, the required anti derivative is the unique function F defined by

$$F(x) = x^4 - 6x + 3$$



Q22

$$\frac{d}{dx}(f(x)) = 4x^3 - \frac{3}{x^4}$$

$$\int \left(4x^3 - \frac{3}{x^4}\right) dx = f(x) + C$$

$$x^4 + \frac{1}{x^3} = f(x) + C$$

Put $x = 2$

$$\frac{16}{1} + \frac{1}{8} = f(2) + C$$

$$\frac{128+1}{8} = 0 + C$$

$$C = \frac{129}{8}$$

21. The anti derivative of $\left(\sqrt{x} + \frac{1}{\sqrt{x}}\right)$ equals

(A) $\frac{1}{3}x^{\frac{1}{3}} + 2x^{\frac{1}{2}} + C$

(B) $\frac{2}{3}x^{\frac{2}{3}} + \frac{1}{2}x^2 + C$

(C) $\frac{2}{3}x^{\frac{3}{2}} + 2x^{\frac{1}{2}} + C$

(D) $\frac{3}{2}x^{\frac{3}{2}} + \frac{1}{2}x^{\frac{1}{2}} + C$

22. If $\frac{d}{dx}f(x) = 4x^3 - \frac{3}{x^4}$ such that $f(2) = 0$. Then $f(x)$ is

(A) $x^4 + \frac{1}{x^3} - \frac{129}{8}$

(B) $x^3 + \frac{1}{x^4} + \frac{129}{8}$

(C) $x^4 + \frac{1}{x^3} + \frac{129}{8}$

(D) $x^3 + \frac{1}{x^4} - \frac{129}{8}$

$$x^4 + \frac{1}{x^3} - \frac{129}{8} = f(x)$$

7.3 Methods of Integration

In previous section, we discussed integrals of those functions which were readily obtainable from derivatives of some functions. It was based on inspection, i.e., on the search of a function F whose derivative is f which led us to the integral of f . However, this method, which depends on inspection, is not very suitable for many functions. Hence, we need to develop additional techniques or methods for finding the integrals by reducing them into standard forms. Prominent among them are methods based on:

1. Integration by Substitution
2. Integration using Partial Fractions
3. Integration by Parts

7.3.1 Integration by substitution

In this section, we consider the method of integration by substitution.

The given integral $\int f(x) dx$ can be transformed into another form by changing the independent variable x to t by substituting $x = g(t)$.

H.W

EX-6.1

Q8 to Q22

Give Each question a max of 3 attempts

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